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A conservative staggered discontinuous Galerkin method on triangular grids for the shallow water equations

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ABSTRACT

In this paper, we present a novel staggered discontinuous Galerkin (SDG) method on general staggered triangular grids for solving the shallow water equations arising in planetary geophysical flows. In the proposed spatial discretization with SDG, the normal component of the velocity function is designed to be continuous in each primal macro element and the fluid height continuous in each dual macro element. Our SDG method preserves key conservation properties, simultaneously ensuring local conservation of the mass and global conservation of the energy and vorticity. Dispersion analysis also reveals that our SDG method is suitable to suppress the potential spurious modes by maintaining an appropriate ratio between the degrees of freedom of the vorticity, velocity and fluid height variables. By further employing our SDG scheme alongside the mass-lumping finite element method for the vorticity equation, block diagonal mass matrices for velocity and height equations and diagonal mass matrices for vorticity equation are achieved, which can facilitate the inversion of mass matrices and consequently enhance the computational efficiency of explicit time stepping. Several benchmark experiments are also carried out to validate the proposed SDG method and demonstrate its conservation properties and efficiency.

1. Introduction

Traditional atmospheric and ocean models using a latitude-longitude grid have encountered computational bottlenecks due to meridian convergence at the poles [1]. To address this issue, unstructured grids without poles have been employed, meeting the scalability requirements of climate and ocean modeling, which aims for exascale computing. Additionally, ocean models employing variable resolution meshes have been commonly applied in coastal, regional, and global simulations [2]. Compared to the nested structured latitude-longitude grids, unstructured grids can support variable resolution more easily and consistently, allowing for the seamless integration of local models within global models.

Compatible finite element methods on quadrilateral and triangular grids have been quite successful in numerical weather prediction [3,4], serving as an extension to the Arakawa C-grid staggered finite difference methods on the latitude-longitude grid [5]. In the C-grid discretization, different physical fields such as the velocity components and the pressure are staggered across different grid entities, effectively mitigating issues associated with spurious waves. In the pioneering work of Cotter and Shipton, and subsequent co-workers [6–8], a hierarchy of finite element methods (FEMs) function spaces are utilized for different fields, and the discretized vector calculus operators of these spaces are designed such that both the curl of gradient and the divergence of curl remain identically

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zero and the commuting diagram enables a discrete Helmholtz decomposition for the velocity field. The numerical method developed from this approach preserves both first and second-order momentum conservation, as well as geostrophic modes, while effectively suppressing the spurious modes associated with inertial-gravity and Rossby waves. The issue of spurious modes in the discretization of the nonlinear shallow water equations was identified in [9], and numerous finite element methods have been analyzed to address this problem [10,11]. It has been found that the spurious modes are closely related to the dimensional relationship between the vorticity, velocity, and height variables [6], providing important guidelines for selecting appropriate FEM spaces for compatible FEM discretization to follow. These advancements have led to further development of the compatible FEMs as dynamical core of numerical weather prediction models [4]. High order compatible FEMs with quadrilateral elements were also developed for shallow water equations on cubed-sphere grid [12]. Mimetic and compatible discretization schemes based on the finite volume method [2,13] have also focused on conservation of some key physical properties and reduction of numerical dispersion, further advancing numerical predictions in weather forecast and ocean circulation.

Although the compatible FEMs for the shallow water equations possess desired conservation properties in space, they often encounter the efficiency challenges for time stepping. With implicit time stepping schemes, each time step requires solving a full nonlinear equation. When explicit time stepping schemes are used, linear system involving the mass matrix needs to be solved with certain time step size restriction at each time step. In comparison, the Arakawa C-grid discretization on latitude-longitude grids with explicit stepping schemes only need to perform pointwise value updates, which are much more efficient. The lowest order compatible FEM, such as RT0/DG0, has a non-diagonal mass matrix using exact Gauss-Legendre quadrature, however, on equilateral triangular grids, it allows for the use of a lumped diagonal matrix while enhancing the accuracy to second order, which has contributed to its popularity [3,4]. To improve the efficiency of explicit time stepping of FEM discretizations for time-domain wave equations, the mass lumping technique and the discontinuous Galerkin (DG) method are often preferred due to their ability to produce diagonal and block diagonal mass matrices, respectively. The spectral-element method, which applies mass lumping on quadrilateral or hexagonal meshes has been widely utilized [14]. For triangular or tetrahedral meshes, constructing mass lumping FEM approximations involves the identification of special quadrature rules and the use of specialized FEM bases, we refer to [15–17] for further details. The DG method naturally produces block diagonal mass matrices organized on an element-wise basis, and also provides considerable flexibility since it admits meshes of various element types, thus it has been popularly applied to wave equations in the first and second order form [18–23].

In this work, we will develop a novel numerical method that combines the staggered discontinuous Galerkin (SDG) approach [20,24] with the mass lumping technique on general staggered triangular grids for the shallow water equations, aimed at achieving the simultaneous conservation of multiple important physical properties and computational efficiency. The SDG method was originally devised for solving acoustic wave equations with energy preservation, and later extended to handle Maxwell and elastic wave equations [25,26]. The method uses unstructured triangular grids derived from polygonal grids, with the function spaces for variables designed to be discontinuous in a staggered fashion. The SDG method for the shallow water equations was first proposed in [27], which uses function spaces and a formulation that differ from those in [20,24] and has demonstrated effective shock-handling capabilities. This method was extended to unstructured triangular grids in [28], and later combined with a posteriori sub-cell finite volume limiters in [29] to enhance robustness near discontinuities. In our proposed scheme, we employ the SDG discretization with FEM function spaces as in [24] for the height and velocity equations of the shallow water equations to ensure local mass and global energy conservation, and the scheme benefits from a block diagonal mass matrix which enhances the efficiency of explicit time stepping. Moreover, the mass lumping FEM approach is applied for the vorticity equation, and consequently the mass matrices in explicit time-stepping schemes are diagonal, rendering an efficient scheme for time marching. The properties of our scheme are carefully analyzed in detail. For the linearized shallow water equations, it is demonstrated that our SDG discretization is absent of spurious pressure modes and allows for a steady geostrophic mode. Furthermore, numerical dispersion analysis shows that our choices of the function spaces for the velocity and height variables are effective in avoiding spurious modes on both f-plane and β-plane. In the context of nonlinear shallow water equations, we have shown that our discretization for the vorticity equation is diagnostic for the main part but prognostic for bubble functions within each element, providing a robust framework for modeling complex dynamic behaviors. As a finite element method, the proposed approach is not a fully mimetic FEM [8], in the sense that it lacks discrete potential enstrophy conservation and has a partially diagnostic vorticity, however it is more efficient for use in explicit time stepping.

The rest of the paper is organized as follows. In Section 2, the SDG method for discretizing the shallow water equations in space is presented and the mass-lumping finite element method for the vorticity equation is introduced to further enhance the computational efficiency through achieving diagonal mass matrices. In Section 3, the local conservation of mass and the global conservation of energy and vorticity are derived, it is shown the proposed method is free of spurious pressure modes and can maintain a steady geostrophic mode. In Section 4, numerical dispersion analysis for the linearized shallow water equations is carried out. In Section 5, we validate the proposed SDG method and demonstrate its performance using some standard benchmark tests, including the Williamson test cases 2 and 5, the barotropically unstable jet flow described by Galewsky, and a regularized dam break. Finally, some concluding remarks are given in Section 6.

2. The SDG method for the shallow water equations

On a two-dimensional surface Ω embedded in three dimensions, the inviscid nonlinear shallow-water equations in the vector invariant form are given by

$$\begin{cases} \dfrac{\partial h}{\partial t} + \nabla \cdot (\mathbf{u}h) = 0. & (1) \\ \dfrac{\partial \mathbf{u}}{\partial t} + (\zeta + f)\mathbf{u}^{\perp} + \nabla\Big(g(h+b) + \dfrac{1}{2}|\mathbf{u}|^2\Big) = 0, & (2) \end{cases}$$

where $\mathbf{u}$ is the two-dimensional surface velocity vector field, h is the fluid height, $\mathbf{k}$ denotes the unit normal vector to the surface, $\zeta = \nabla^{\perp} \cdot \mathbf{u} := (\mathbf{k} \times \nabla) \cdot \mathbf{u}$ is the vorticity, $\mathbf{u}^{\perp} := \mathbf{k} \times \mathbf{u}$, f is the Coriolis parameter, g is the gravitational acceleration, b is the bottom topography, and the gradient and divergence operators are defined intrinsically on the surface. For simplicity, in this paper we assume the shallow-water equations are defined on a surface without boundaries, e.g. the periodic or sphere surface.

The shallow water equations (1)–(2) conserve mass, energy, vorticity and potential enstrophy [30]. The local conservation of the fluid mass is directly guaranteed by the continuity Eq. (1). The energy of the fluid over the whole domain is given by

$$E = \int_{\Omega} \Big[\frac{h}{2}|\mathbf{u}|^2 + gh(\frac{h}{2} + b)\Big] \,\mathrm{dS},$$

where dS is the differential surface area in the surface integral, and its global conservation can be shown by using (1) and (2):

$$\begin{aligned} &\frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega} \Big[\frac{h}{2}|\mathbf{u}|^2 + gh(\frac{h}{2} + b)\Big] \,\mathrm{dS} \\ &\quad = \int_{\Omega} \Big[h \frac{\partial}{\partial t}\Big(\frac{|\mathbf{u}|^2}{2}\Big) + \Big(\frac{|\mathbf{u}|^2}{2} + g(h+b)\Big)\frac{\partial h}{\partial t}\Big] \,\mathrm{dS} \\ &\quad = \int_{\Omega} \Big[-h\mathbf{u} \cdot \nabla\Big(\frac{|\mathbf{u}|^2}{2} + g(h+b)\Big) - \Big(\frac{|\mathbf{u}|^2}{2} + g(h+b)\Big)\nabla \cdot (h\mathbf{u})\Big] \,\mathrm{dS} \\ &\quad = -\int_{\Omega} \nabla \cdot \Big[h\mathbf{u}\Big(\frac{|\mathbf{u}|^2}{2} + g(h+b)\Big)\Big] \,\mathrm{dS} \\ &\quad = 0. \end{aligned} \tag{3}$$

Here we have used the fact that $\mathbf{u}^{\perp} \cdot \mathbf{u} = 0$ and the condition that the surface has no boundaries.

The potential vorticity is defined as

$$q = \frac{\zeta + f}{h}, \tag{4}$$

which plays a crucial role in understanding the dynamics of atmospheric and oceanic flows. By applying the curl operator $\nabla^{\perp}\cdot$ to (2), and notice that $\nabla^{\perp} \cdot \nabla = 0$, we have

$$\frac{\partial}{\partial t}(qh) + \nabla^{\perp} \cdot (\mathbf{u}^{\perp} qh) = 0,$$

therefore,

$$\frac{\partial}{\partial t}(qh) + \nabla \cdot (\mathbf{u}qh) = 0, \tag{5}$$

which is the local conservation law satisfied by the total vorticity. Using (5) and (1), we get

$$\frac{\partial q}{\partial t} + (\mathbf{u} \cdot \nabla)q = 0, \tag{6}$$

which implies the PV remains constant along the trajectories of fluid parcels. In this paper, we refer to the conservation of vorticity as the total potential conservation as (5), rather than the material conservation (6).

The potential enstrophy of the fluid flow over the whole domain is defined to be

$$Z = \int_{\Omega} q^2 h \,\mathrm{dS}$$

and its global conservation can be obtained by using (2) and (5):

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega} q^2 h \,\mathrm{dS} &= \int_{\Omega} \Big[2q\frac{\partial}{\partial t}(qh) - q^2 \frac{\partial h}{\partial t}\Big] \,\mathrm{dS} \\ &= -\int_{\Omega} \big[2q\nabla \cdot (qh\mathbf{u}) - q^2 \nabla \cdot (h\mathbf{u})\big] \,\mathrm{dS} \\ &= -\int_{\Omega} \nabla \cdot \big(q^2 h\mathbf{u}\big) \,\mathrm{dS} = 0. \end{aligned} \tag{7}$$

2.1. Staggered triangular grids and SDG discretization for velocity and height

Let us consider the case of Ω being a sphere surface for illustration. The SDG discretization in space utilizes special staggered meshes. Starting with a polygonal mesh covering the surface of a sphere, each polygon is then subdivided into multiple triangles by adding an interior vertex, creating a new mesh referred to as $\mathcal{T}_h$. Each polygon corresponds to a single primal macro element (ME),

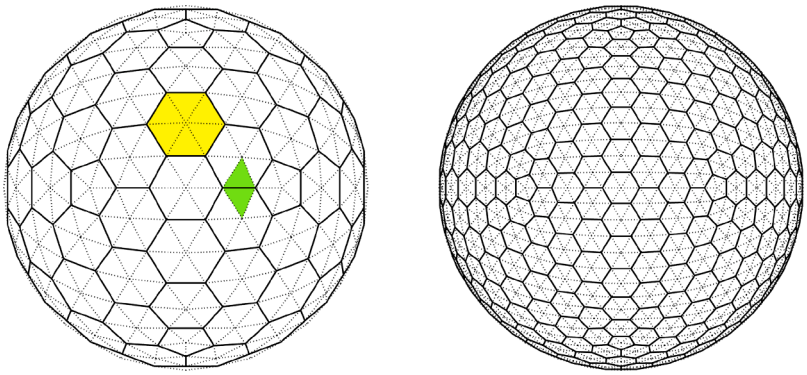


Fig. 1. An illustration of two SDG grids on the sphere with different resolutions. In the left grid, a yellow hexagon represents one primal ME, while a green quadrilateral represents one dual ME. The exterior edges of the primal MEs ($\mathcal{E}_{\mathrm{H1}}$) are represented with solid lines, while the interior edges ($\mathcal{E}_{\mathrm{Hdiv}}$) are represented with dotted lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

which consists of all the triangles contained within it, and we denote the set of all primal MEs as $\mathcal{M}_{\mathrm{Prml}}$. The two neighboring triangles associated with each polygon edge form one dual ME and we denote the set of all dual MEs as $\mathcal{M}_{\mathrm{Dual}}$. The primal and dual MEs are staggered, in the sense that each primal ME is composed of parts of several dual MEs, and vice versa. The set of all the exterior edges of the primal MEs is denoted as $\mathcal{E}_{\mathrm{H1}}$, which is also the set of interior edges of the dual MEs. The set of all the interior edges of the primal MEs is denoted as $\mathcal{E}_{\mathrm{Hdiv}}$, which is also the set of exterior edges of the dual MEs. The edges in $\mathcal{E}_{\mathrm{H1}}$ and $\mathcal{E}_{\mathrm{Hdiv}}$ constitute all the edges of the triangular mesh $\mathcal{T}_h$, which we denote by $\mathcal{E}$. An illustration of the SDG grids with primal and dual MEs on the sphere is shown in Fig. 1.

We use a nonlinear element mapping to define finite element spaces on the sphere. Regarding the nonlinear coordinate mapping and the construction of mixed FEM spaces on the sphere we refer to [31] for details. For any triangle $\tau \in \mathcal{T}_h$, there is an affine mapping from the 2D reference triangle $\hat{\tau}$ to the flat triangle τ on the sphere, which is denoted by $F_{\tau,0}(\hat{\mathbf{x}})$ for any point $\hat{\mathbf{x}} \in \hat{\tau}$. Then with the projection function $P(\mathbf{x}) = R\frac{\mathbf{x}}{\|\mathbf{x}\|}$, where R is the sphere radius, the nonlinear element mapping is defined as $F_\tau = P \circ F_{\tau,0}$, such a nonlinear mapping maps from the reference triangle $\hat{\tau}$ to a curved triangle $\widetilde{\tau}$ on the sphere. Denote DF_τ as the Jacobian matrix of the transformation $F_\tau(\hat{\mathbf{x}})$, which could be written as $DF_\tau = (DF_{\tau,1}, DF_{\tau,2})$, where $\{DF_{\tau,i}\}_{i=1}^2$ are 3D vectors depending on $\hat{\mathbf{x}}$. The determinant of DF_τ is denoted as

$$J_\tau = \|DF_{\tau,1} \times DF_{\tau,2}\|.$$

The finite element spaces for our SDG discretization are then constructed as follows. Denote $P_k(\hat{\tau})$ as the set of piecewise polynomials of degree k in the triangle $\hat{\tau}$. In each primal ME, the normal component of the vector function is required to be continuous on its interior edges, resulting in a local $H(\mathrm{div})$ conforming finite element space, defined as

$$\mathbb{U}_h = \left\{ \mathbf{v}_h \,\middle|\, \mathbf{v}_h|_\tau = \frac{1}{J_\tau} DF_\tau \hat{\mathbf{v}} \circ F_\tau^{-1} \text{ with } \hat{\mathbf{v}} \in (P_k(\hat{\tau}))^2 \text{ for any } \tau \in \mathcal{T}_h, \text{ and } \mathbf{v}_h \cdot \mathbf{n} \text{ is continuous on any } e \in \mathcal{E}_{\mathrm{Hdiv}} \right\}, \tag{8}$$

where $\mathbf{n}$ on each $e \in \mathcal{E}_{\mathrm{Hdiv}}$ is defined as the unit normal counterclockwise around the corresponding ME. In each dual ME, the continuity of the scalar function is required on its interior edge, resulting in a local H_1 conforming finite element space, defined as

$$\mathbb{V}_h = \left\{ \phi_h \,\middle|\, \phi_h|_\tau \circ F_\tau \in P_k(\hat{\tau}) \text{ for any } \tau \in \mathcal{T}_h, \text{ and } \phi_h \text{ is continuous on any } e \in \mathcal{E}_{\mathrm{H1}} \right\}. \tag{9}$$

To illustrate the SDG basis functions, the positions of the Lagrangian type basis functions for the first order SDG finite element spaces $\mathbb{U}_h$ and $\mathbb{V}_h$ are shown in Fig. 2. Specifically, for $\mathbb{U}_h$, two basis functions are associated with each edge of a triangle, located at the two Gauss quadrature points on that edge. On the reference triangle $\hat{\tau}$, let $\{\hat{\mathbf{x}}_j\}_{j=1}^6$ denote the positions of six evaluation nodes and $\{\hat{\mathbf{n}}_j\}_{j=1}^6$ be the corresponding outward unit normal vectors. Then the six vector valued basis functions $\hat{\psi}_i$ on the reference element are taken in $(P_1(\hat{\tau}))^2$ and defined by the interpolation conditions

$$\hat{\psi}_i(\hat{\mathbf{x}}_j) \cdot \hat{\mathbf{n}}_j = \delta_{ij}, \quad \text{for } i, j = 1, \ldots, 6.$$

Finally, with the contravariant Piola transformation, the vector bases $\hat{\psi}_i$ are transformed back to the physical elements by $\psi_i = \frac{1}{J_\tau} DF_\tau \hat{\psi}_i$ to constitute the basis functions of $\mathbb{U}_h$. These basis functions coincide with the BDM_1 basis functions, however, we create separate copies on each primal ME, yielding discontinuity across primal macro element boundaries. For $\mathbb{V}_h$, one basis function is associated with each vertex of the triangle like the standard piecewise linear functions which is enforced to be continuous inside each dual ME. Separate copies are again created on each dual ME and thus they are discontinuous across dual MEs. The direction and relative magnitudes of the first order vector basis functions of $\mathbb{U}_h$ are correspondingly depicted in Fig. 3.

Using the divergence theorem and notice that $\mathcal{E}_{\mathrm{H1}} \cup \mathcal{E}_{\mathrm{Hdiv}} = \mathcal{E}$, it holds that for any $\mathbf{v}_h \in \mathbb{U}_h$ and $\phi_h \in \mathbb{V}_h$,

$$(\mathbf{v}_h, \nabla \phi_h) - \langle \mathbf{v}_h \cdot \mathbf{n}, [\![\phi_h]\!] \rangle_{\mathcal{E}_{\mathrm{Hdiv}}} = -(\nabla \cdot \mathbf{v}_h, \phi_h) + \langle [\![\mathbf{v}_h \cdot \mathbf{n}]\!], \phi_h \rangle_{\mathcal{E}_{\mathrm{H1}}}. \tag{10}$$

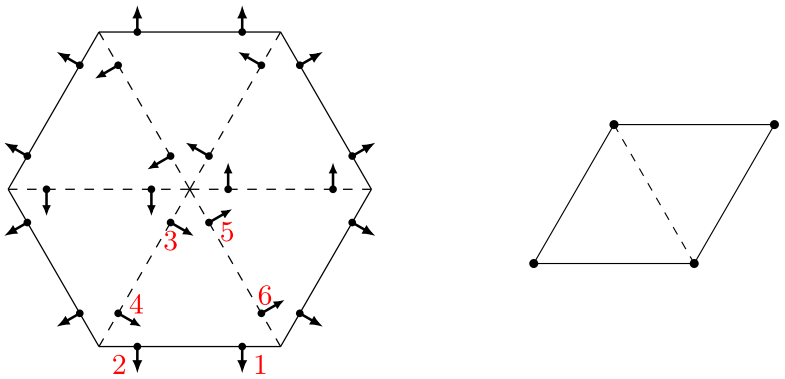


Fig. 2. Illustration of the basis functions for the first order SDG finite element spaces. Left: the nodes of the basis functions for $\mathbb{U}_h$ where the numbers 1–6 indicate the basis function indices in the triangle at the bottom. Right: the nodes of the basis functions for $\mathbb{V}_h$.

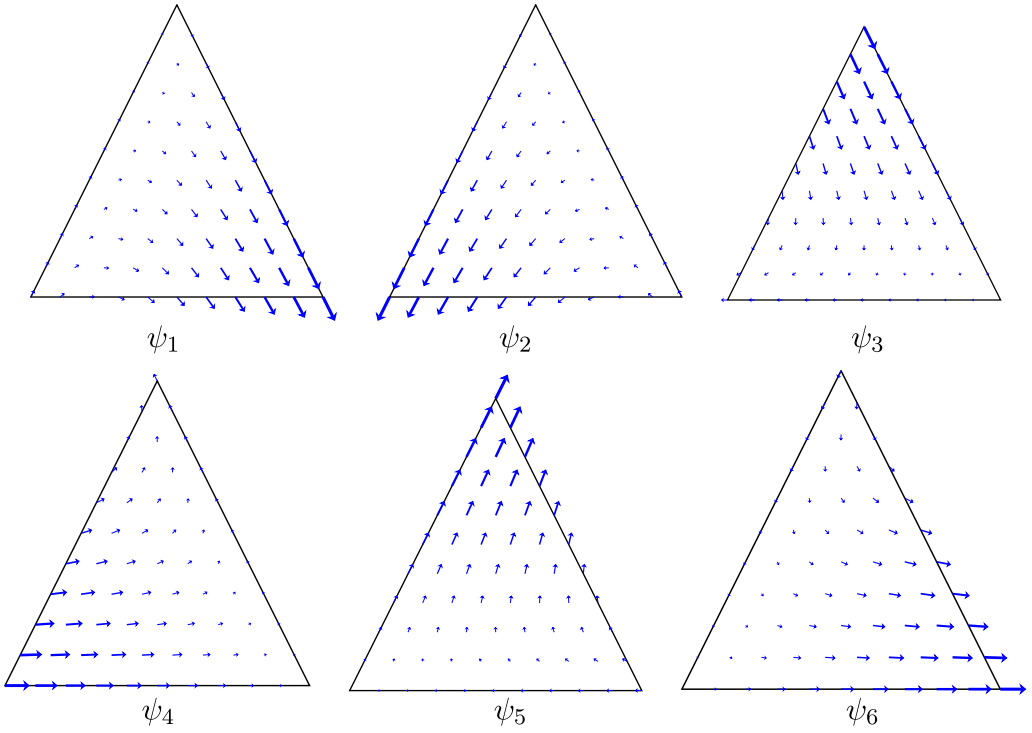


Fig. 3. The directions and relative magnitudes of the six vector basis functions of first order SDG finite element space $\mathbb{U}_h$ corresponding to the triangle marked in the left panel of Fig. 2.

The spatial discretization of the shallow water equations (1)–(2) with the SDG method is then given by: find $\mathbf{u}_h \in \mathbb{U}_h$, $h_h \in \mathbb{V}_h$, $\mathbf{F} \in \mathbb{U}_h$ and $U \in \mathbb{V}_h$ such that

$$
\begin{cases}
\left(\dfrac{\partial h_h}{\partial t}, \phi_h\right) - (\mathbf{F}, \nabla\phi_h) + \langle \mathbf{F}\cdot\mathbf{n}, [\![\phi_h]\!]\rangle_{\mathcal{E}_{\mathrm{Hdiv}}} = 0, & \forall\, \phi_h \in \mathbb{V}_h, \quad (11)\\
\left(\dfrac{\partial \mathbf{u}_h}{\partial t}, \mathbf{v}_h\right) + \left(q_h\mathbf{F}^{\perp}, \mathbf{v}_h\right) - \left(g(h_h+b) + \dfrac{1}{2}U, \nabla\cdot\mathbf{v}_h\right) & \\
\qquad + \left\langle g(h_h+b) + \dfrac{1}{2}U, [\![\mathbf{v}_h\cdot\mathbf{n}]\!]\right\rangle_{\mathcal{E}_{\mathrm{H1}}} = 0, & \forall\, \mathbf{v}_h \in \mathbb{U}_h, \quad (12)\\
(\mathbf{F}, \mathbf{v}_h) = (\mathbf{u}_h h_h, \mathbf{v}_h), & \forall\, \mathbf{v}_h \in \mathbb{U}_h, \quad (13)\\
(U, \phi_h) = (|\mathbf{u}_h|^2, \phi_h), & \forall\, \phi_h \in \mathbb{V}_h, \quad (14)
\end{cases}
$$

where $(\cdot,\cdot)$ denotes the inner product over the domain Ω and $\langle\cdot,\cdot\rangle_{\mathcal{E}}$ denotes the inner product over the line segments $\mathcal{E}$. Note that the discretization of potential vorticity q_h needed in (12) will be described in the next subsection. Since the discrete velocity function is discontinuous across the primal ME, and the discrete height function is discontinuous across the dual ME, the resulting mass matrices from the above discretization are block diagonal.

2.2. Mass lumping finite element method for the vorticity equation

The potential vorticity q is crucial for understanding, predicting, and modeling dynamic patterns in both the ocean and atmosphere, it is commonly treated as a diagnostic variable and derived from velocity using (4). However, the requirement to discretely conserve vorticity, similar to its conservation in continuous formulations as outlined in Eq. (7), necessitates a discrete prognostic equation. Conservative numerical schemes such as [2,6] successfully ensure compatibility between the discrete diagnostic and prognostic equations. Enlightened by the de Rham complex, the finite element space

$$\mathbb{W}_h = \left\{ q_h \,\middle|\, q_h|_\tau \circ F_\tau \in P_{k+1}(\widehat{\tau}), q_h \in H^1(\Omega) \right\}, \tag{15}$$

is a plausible space for discretizing potential vorticity q. The following lemma on subspace properties, analogous to the affine coordinate mapping case, holds for the FEM spaces on the sphere.

Lemma 2.1. *The spaces defined in (8) and (15) satisfy*

$$\nabla^\perp \mathbb{W}_h \subset \mathbb{U}_h, \tag{16}$$

moreover,

$$[\![\mathbf{n}\cdot\nabla^\perp\gamma_h]\!]\Big|_e = 0, \quad \forall \gamma_h \in \mathbb{W}_h, \forall e \in \mathcal{E}. \tag{17}$$

Proof. We first verify that, for any triangle $\tau \in \mathcal{T}_h$, the function $\nabla^\perp\gamma_h|_\tau$ conforms to the representation of the space $\mathbb{W}_h$ in (15) within the element. For simplicity, throughout the remainder of the proof, the subscript τ is omitted, and hatted quantities refer to those defined on the reference element. Let $G = [G_1, G_2]^T$ denote the generalized inverse of the Jacobian matrix DF, which has dimensions 2×3. The orthogonality condition states that $(G_i, DF_j) = \delta_{ij},\ i,j = 1,2$. Note that the unit normal vector is given by

$$\mathbf{k} = \frac{DF_1 \times DF_2}{\|DF_1 \times DF_2\|} = \frac{DF_1 \times DF_2}{J}.$$

For any $\gamma_h \in \mathbb{W}_h$, we then have

$$\begin{aligned}
\mathbf{k}\times\nabla\gamma_h = \mathbf{k}\times(G^T\widehat{\nabla}\widehat{\gamma}) &= \frac{DF_1\times DF_2}{J}\times\left(G_1\,\partial_{\hat{x}}\widehat{\gamma} + G_2\,\partial_{\hat{y}}\widehat{\gamma}\right)\\
&= -\frac{1}{J}\left((G_1, DF_2)DF_1 - (G_1, DF_1)DF_2\right)\partial_{\hat{x}}\widehat{\gamma}\\
&\quad -\frac{1}{J}\left((G_2, DF_2)DF_1 - (G_2, DF_1)DF_2\right)\partial_{\hat{y}}\widehat{\gamma}\\
&= \frac{1}{J}\left(DF_2\,\partial_{\hat{x}}\widehat{\gamma} - DF_1\,\partial_{\hat{y}}\widehat{\gamma}\right)\\
&= \frac{1}{J}DF(\widehat{\mathbf{k}}\times\widehat{\nabla}\widehat{\gamma}).
\end{aligned}$$

and this expression conforms to the representation of the space $\mathbb{W}_h$ in (15) within the element.

Next notice that $\gamma_h \in \mathbb{W}_h$ is continuous across the sharing edge of any neighboring elements, thus its tangential derivative along the edge is also continuous across edges, then we have

$$[\![\mathbf{n}\cdot\nabla^\perp\gamma_h]\!]|_e = [\![\mathbf{t}\cdot\nabla\gamma_h]\!]|_e = 0, \quad \forall e \in \mathcal{E}, \tag{18}$$

where $\mathbf{t}$ is the tangential direction on the edge. Therefore the normal continuity condition in (15) is also satisfied and the lemma follows.

The curl of the velocity is defined in a distributed sense as in the compatible FEM [6]. Specifically, the discrete PV is determined by finding $\widetilde{q}_h \in \mathbb{W}_h$ such that

$$\left(\widetilde{q}_h h_h, \gamma_h\right) + \left(\mathbf{u}_h, \nabla^\perp\gamma_h\right) - (f, \gamma_h) = 0, \qquad \forall\,\gamma_h \in \mathbb{W}_h, \tag{19}$$

using this diagnostic equation together with (12), the prognostic form of the vorticity equation could be deduced.

However, the corresponding mass matrix of the space $\mathbb{W}_h$ is not diagonal, and directly applying mass lumping on the mass matrix will degrade the accuracy [15]. The non-diagonal mass matrix need an iterative solver for solution of the discrete system of vorticity equation, which slows down the computation efficiency when compared to the block diagonal mass solve in discontinuous Galerkin schemes for velocity and depth. Hence we will construct a finite element space with bubbles that contains $\mathbb{W}_h$ and also supports mass lumping, which makes solving the mass matrix system more efficient. On the other hand, the resulting SDG scheme could make discrete vorticity no longer completely diagnostic, but partial prognostic for the additional bubble bases, as we explain in the following.

The mass lumping finite element space on triangles [15] enriches the Lagrangian space P_{k+1} with additional bubbles, for example, the second order mass lumping finite element space is given by

$$\widetilde{P}_2 = P_2 \oplus \mathcal{B} \tag{20}$$

where $b = \lambda_1\lambda_2\lambda_3$ is the bubble function with λ_i, $i = 1, 2, 3$ being the barycentric coordinates with respect to the three vertices of the triangle. The third order mass lumping finite element space is given by

$$\widetilde{P}_3 = P_3 \oplus \mathcal{B}\overline{P}_1, \tag{21}$$

where $\overline{P}_1$ denote homogeneous polynomial of order one. Higher order mass lumping finite element spaces could be found in [15,17]. Thus the $(k+1)$-th order mass lumping finite element space for vorticity is taken as

$$\widetilde{\mathbb{W}}_h = \left\{ q_h \,\middle|\, q_h|_\tau \circ F_\tau \in \widetilde{P}_{k+1}(\hat{\tau}), q_h \in H^1(\Omega) \right\}, \tag{22}$$

corresponding to the k-th order SDG spaces $\mathbb{U}_h$ and $\mathbb{V}_h$. It is obvious that $\mathbb{W}_h$ is a subspace of $\widetilde{\mathbb{W}}_h$ and $\widetilde{\mathbb{W}}_h \setminus \mathbb{W}_h$ is the subspace composed of higher order additional bubble functions in each of the triangles. The vorticity Eq. (5) is then discretized as follows: find $q_h \in \widetilde{\mathbb{W}}_h$, such that

$$\begin{cases} \left(q_h h_h, \gamma_h\right)_{\mathrm{ML}} + \left(\mathbf{u}_h, \nabla^\perp \gamma_h\right) - \left(f, \gamma_h\right) = 0, & \forall\, \gamma_h \in \mathbb{W}_h, \quad (23) \\ \left(\partial_t (q_h h_h), \gamma_h\right)_{\mathrm{ML}} - \left(q_h \mathbf{F}, \nabla \gamma_h\right) = 0, & \forall\, \gamma_h \in \widetilde{\mathbb{W}}_h \setminus \mathbb{W}_h, \quad (24) \end{cases}$$

where the mass-related inner products $(\cdot,\cdot)_{\mathrm{ML}}$ are particularly computed using mass lumping quadrature rules associated with the mass lumping bases in each element, which lead to a diagonal mass matrix without compromising the accuracy. As in (19), the Eq. (23) relates the vorticity $q_h h_h \in \widetilde{\mathbb{W}}_h$ to the weak curl of velocity $\mathbf{u}_h \in \mathbb{U}_h$, which forms the diagnostic part of the vorticity. Note the difference between (19) and (23) is that the vorticity is defined in different spaces. On the other hand, the prognostic equation (24) with bubble testing functions controls the prognostic part of the vorticity. From the perspective of function spaces, the diagnostic part constitutes the main component of vorticity, consisting of the function that is continuous and piecewise k-th order polynomial. In contrast, the prognostic part is a smaller contribution, comprising only the additional bubble functions within each element. In such sense, the deviation from diagnostic vorticity is considered mild.

Remark 2.2. An alternative choice for the vorticity space is $\mathbb{W}_h$ in (15). In this formulation, the potential vorticity is defined by (19), and the conservation properties of mass, energy, and vorticity can be derived similarly. This approach ensures that the vorticity remains fully diagnostic; however, it results in a non-local mass matrix for vorticity. Thus, there is a trade-off between maintaining vorticity as a purely diagnostic quantity with a non-local mass matrix and using a partially prognostic formulation with a local mass matrix. In this paper, we focus on time-stepping efficiency and therefore consider the latter one as a complementary choice.

Remark 2.3. We note that the vorticity space (20) is also used in the $(\mathrm{P}_2 \oplus \mathcal{B}, \mathrm{BDFM}_1, \mathrm{DG}_1)$ compatible FEM scheme [6]. This scheme maintains an optimal ratio of degrees of freedom for velocity and height, making it well-suited for reducing spurious inertial-gravity and Rossby oscillations. Unlike our proposed SDG scheme, this approach fully preserves diagnostic vorticity.

The system of Eqs. (11)–(14) and (23)–(24) forms our complete discretization scheme for solving the shallow water equations (1)–(2). For convenience, we refer to the above numerical scheme with the k-th order SDG spaces for velocity and thickness and the $(k+1)$-th order mass lumping space for vorticity as the SDG_k scheme.

3. Important properties of the SDG method

Now we derive and show some important conservation and stability properties for the proposed SDG method.

3.1. Mass and energy conservation

For computational atmosphere and ocean models, the local mass conservation is a necessary requirement, while the global energy conservation is a desirable property, we have the following theorem of conservation.

Theorem 3.1. *The SDG discretization scheme (11)–(14) satisfies local mass conservation on primal macro elements and global energy conservation.*

Proof. On each primal macro element $\mathcal{M}_{\mathrm{Prml}}$, by taking ϕ to be the characteristic function on $\mathcal{M}_{\mathrm{Prml}}$ (i.e., $\phi_h = \chi_{\mathcal{M}_{\mathrm{Prml}}}$, so that $\phi_h = 1$ in $\mathcal{M}_{\mathrm{Prml}}$ and zero elsewhere) in (11), it shows

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathcal{M}_{\mathrm{Prml}}} h_h \,\mathrm{dS} &= \left(\mathbf{F}, \nabla \chi_{\mathcal{M}_{\mathrm{Prml}}}\right) - \left\langle \mathbf{F}\cdot\mathbf{n}, [\![\chi_{\mathcal{M}_{\mathrm{Prml}}}]\!] \right\rangle_{\mathcal{E}_{\mathrm{Hdiv}}} \\ &= -\int_{\mathcal{M}_{\mathrm{Prml}}} \nabla \cdot \mathbf{F} \,\mathrm{dS} \\ &= -\int_{\partial \mathcal{M}_{\mathrm{Prml}}} \mathbf{F}\cdot\mathbf{n} \,\mathrm{d}\sigma. \end{aligned} \tag{25}$$

where $\mathrm{d}\sigma$ is the differential line element in the line integral.

Thus we have local mass conservation on primal macro elements.

As for the energy, using the definition of $\mathbf{F}$ and U in (13) and (14), we have

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega\left(\frac{h_h}{2}|\mathbf{u}_h|^2+gh_h(\frac{h_h}{2}+b)\right)\mathrm{dS}=&\left(\frac{\partial\mathbf{u}_h}{\partial t},\mathbf{u}_hh_h\right)+\left(\frac{\partial h_h}{\partial t},g(h_h+b)+\frac{|\mathbf{u}_h|^2}{2}\right)\\=&\left(\frac{\partial\mathbf{u}_h}{\partial t},\mathbf{F}\right)+\left(\frac{\partial h_h}{\partial t},g(h_h+b)+\frac{1}{2}U\right).\end{aligned}\tag{26}$$

Taking $\mathbf{v}_h=\mathbf{F}$ in (12), and notice $\left(q\mathbf{F}^\perp,\mathbf{F}\right)=0$, then we have

$$\left(\frac{\partial\mathbf{u}_h}{\partial t},\mathbf{F}\right)-\left(g(h_h+b)+\frac{1}{2}U,\nabla\cdot\mathbf{F}\right)+\left\langle g(h_h+b)+\frac{1}{2}U,[\![\mathbf{F}\cdot\mathbf{n}]\!]\right\rangle_{\mathcal{E}_{\mathrm{H1}}}=0.\tag{27}$$

Taking $\phi_h=g(h_h+b)+\frac{1}{2}U$ in (11), then we have

$$\begin{aligned}&\left(\frac{\partial h_h}{\partial t},g(h_h+b)+\frac{1}{2}U\right)-\left(\mathbf{F},\nabla\left(g(h_h+b)+\frac{U}{2}\right)\right)\\&\qquad+\left\langle\mathbf{F}\cdot\mathbf{n},[\![g(h_h+b)+\frac{1}{2}U]\!]\right\rangle_{\mathcal{E}_{\mathrm{Hdiv}}}=0.\end{aligned}\tag{28}$$

Combining (26)–(28), with divergence theorem (10), we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega\left(\frac{h_h}{2}|\mathbf{u}_h|^2+gh_h(\frac{h_h}{2}+b)\right)\mathrm{dS}=0,\tag{29}$$

which gives the global conservation of energy.

The proposed scheme conserves discrete mass and energy; however, discrete potential enstrophy is not conserved, as will be explained in a later subsection.

3.2. Vorticity conservation

The Eq. (23) shows that q_hh_h is diagnostic in the space $\mathbb{W}_h$, which serves as the primal subspace of $\widetilde{\mathbb{W}}_h$, on the other hand, the Eq. (24) shows that q_hh_h is prognostic in the bubble subspace of $\widetilde{\mathbb{W}}_h$. The two Eqs. (23)–(24) are compatible with a single prognostic form, leading us to the following theorem.

Theorem 3.2. *The vorticity discretization scheme (23)–(24) satisfies the prognostic equation*

$$\left(\frac{\partial(q_hh_h)}{\partial t},\gamma_h\right)_{ML}-(q_h\mathbf{F},\nabla\gamma_h)=0,\qquad\forall\gamma_h\in\widetilde{\mathbb{W}}_h,\tag{30}$$

and the vorticity q_h satisfies global conservation.

Proof. Taking $\gamma_h\in\mathbb{W}_h$, by (19) we can use $\nabla^\perp\gamma_h$ as a test function in (12), together with (17) and $\nabla\cdot\nabla^\perp\gamma_h=0$, and we obtain

$$\left(\frac{\partial\mathbf{u}_h}{\partial t},\nabla^\perp\gamma_h\right)_{\mathrm{ML}}+\left(q\mathbf{F}^\perp,\nabla^\perp\gamma_h\right)=0,\qquad\forall\gamma_h\in\mathbb{W}_h.\tag{31}$$

Together with the diagnostic Eq. (23), we have

$$\left(\frac{\partial(q_hh_h)}{\partial t},\gamma_h\right)_{\mathrm{ML}}-(q_h\mathbf{F},\nabla\gamma_h)=0,\qquad\forall\gamma_h\in\mathbb{W}_h,\tag{32}$$

Combining (32) with (24), we have a prognostic equation in the whole space $\widetilde{\mathbb{W}}_h$ as in (30).

The global conservation of vorticity is then easily obtained by choosing the testing function $\gamma_h=1$ in (32),

$$\frac{d}{dt}\int_{\Omega,\mathrm{ML}}(q_hh_h)\,\mathrm{dS}=0,\tag{33}$$

where the integral is computed using the mass lumping quadrature elementwise.

Remark 3.3. In the alternative choice of vorticity in $\mathbb{W}_h$, the conservation of vorticity could be proved similarly. Taking $\nabla^\perp\gamma_h$, $\gamma_h\in\mathbb{W}_h$ as the testing function and using diagnostic definition of vorticity (19), we have

$$\left(\frac{\partial(q_hh_h)}{\partial t},\gamma_h\right)-(q_h\mathbf{F},\nabla\gamma_h)=0,\qquad\forall\gamma_h\in\mathbb{W}_h,$$

and the vorticity conservation follows by taking $\gamma_h=1$.

3.3. Absence of spurious pressure modes

The collocated discretization that is analogous to the A-grid discretization often suffers from checkerboard pressure modes. On A-grids this mode allows for oscillatory numerical pressure with zero discrete gradient, despite having a non-zero actual gradient. For the collocated discretization on unstructured mesh, this mode often leads to oscillatory numerical pressure for which the discrete gradient tends to zero as the grid refines, yet its norm is bounded above zero. Such spurious pressure modes can rapidly pollute the numerical solution, however the staggered discretizations on C-grids are effective in eliminating these artifacts, although they may still exhibit other computational modes depending on the ratio of velocity to pressure degrees of freedom.

In our SDG discretization, the discrete gradient as in (12) is a linear operator from any $\phi_h \in \mathbb{V}_h$ to $\mathbb{U}_h$, defined as

$$(\mathrm{D_h}\phi_h, \mathbf{v}_h) = B_h^*(\phi_h, \mathbf{v}_h), \qquad \forall\, \mathbf{v}_h \in \mathbb{U}_h. \tag{34}$$

where

$$B_h^*(\phi_h, \mathbf{v}_h) = -\int_\Omega \phi_h \nabla\cdot\mathbf{v}_h \,\mathrm{dS} + \sum_{e\in\mathcal{E}_{\mathrm{H1}}} \int_\gamma \phi_h [\![\mathbf{v}_h\cdot\mathbf{n}]\!]\,\mathrm{d}\sigma. \tag{35}$$

Based on the result of [24], we have the following inf-sup condition: there is a grid independent constant $\beta > 0$ such that

$$\inf_{\phi_h\in\mathbb{V}_h}\sup_{\mathbf{v}_h\in\mathbb{U}_h} \frac{B_h^\star(\phi_h,\mathbf{v}_h)}{\|\mathbf{v}_h\|_{0,h}|\phi_h|_{1,h}} \geqslant \beta, \tag{36}$$

where

$$|\phi_h|_{1,h}^2 = \int_\Omega |\nabla\phi_h|^2\,\mathrm{dS} + \sum_{\gamma\in\mathcal{E}_{\mathrm{Hdiv}}} \mathrm{diam}(\gamma)^{-1}\int_\gamma [\![\phi_h]\!]^2\,\mathrm{d}\sigma,$$

$$\|\mathbf{v}_h\|_{0,h}^2 = \int_\Omega |\mathbf{v}_h|^2\,\mathrm{dS} + \sum_{\gamma\in\mathcal{E}_{\mathrm{Hdiv}}} \mathrm{diam}(\gamma)\int_\gamma (\mathbf{v}_h\cdot\mathbf{n})^2\,\mathrm{d}\sigma.$$

The above inf-sup condition (36) could be rewritten as

$$\|\mathrm{D_h}\phi_h\| \geqslant \beta|\phi_h|_{1,h}, \tag{37}$$

which indicates that for any $\phi_h \in \mathbb{V}_h$, the norm of its discrete gradient $\mathrm{D_h}\phi_h$ will be bounded below by the norm of its true gradient $\nabla\phi_h$, hence there will be no spurious pressure modes in the proposed SDG discretization.

3.4. The Helmholtz decomposition

The analysis of discretization schemes for the shallow water equations is typically conducted on the linearized shallow water equations. This allows for verifying the existence of desirable physical modes and assessing the presence and characteristics of spurious modes. The linearized shallow water equations are given by:

$$\begin{cases} \dfrac{\partial h}{\partial t} + H\nabla\cdot\mathbf{u} = 0, & (38)\\ \dfrac{\partial \mathbf{u}}{\partial t} + f\mathbf{u}^\perp + g\nabla h = 0, & (39) \end{cases}$$

where H is the mean layer thickness, and the corresponding SDG discretization applied to (38)–(39) becomes

$$\begin{cases} \left(\dfrac{\partial h_h}{\partial t}, \phi_h\right) - (\mathbf{u}_h H, \nabla\phi_h) + \langle \mathbf{u}_h H\cdot\mathbf{n}, [\![\phi_h]\!]\rangle_{\mathcal{E}_{\mathrm{Hdiv}}} = 0, & \forall\,\phi_h\in\mathbb{V}_h, \quad (40)\\ \left(\dfrac{\partial \mathbf{u}_h}{\partial t}, \mathbf{v}_h\right) + \left(f\mathbf{u}_h^\perp, \mathbf{v}_h\right) - \left(gh_h\nabla\cdot\mathbf{v}_h\right) + \langle gh_h, [\![\mathbf{v}_h\cdot\mathbf{n}]\!]\rangle_{\mathcal{E}_{\mathrm{H1}}} = 0, & \forall\,\mathbf{v}_h\in\mathbb{U}_h. \quad (41) \end{cases}$$

The analysis of discrete linearized shallow water equations (40)–(41) relies on the discrete Helmholtz decomposition. In the SDG scheme, the finite element space $\mathbb{W}_h$ defined as (15) plays an important role in the analysis. Both the space of curl of $\mathbb{W}_h$ and the space of discrete gradient of $\mathbb{V}_h$ are subspaces of $\mathbb{U}_h$, and they are orthogonal since

$$(\nabla^\perp\gamma_h, \mathrm{D_h}\phi_h) = -\left(\nabla\cdot(\nabla^\perp\gamma_h), \phi_h\right) + \left\langle [\![\nabla^\perp\gamma_h\cdot\mathbf{n}]\!], \phi_h\right\rangle_{\mathcal{E}_{\mathrm{H1}}} = 0, \qquad \forall\,\gamma_h\in\mathbb{W}_h,\ \phi_h\in\mathbb{V}_h. \tag{42}$$

Hence we have a discrete Helmholtz decomposition, for any $\mathbf{u}_h \in \mathbb{U}_h$, there exist $\Phi_h \in \mathbb{V}_h$, $\Psi_h \in \mathbb{W}_h$ and $\Theta_h \in \mathbb{H}_h$, such that

$$\mathbf{u}_h = \mathrm{D_h}\Phi_h + \nabla\times\Psi_h + \Theta_h, \tag{43}$$

where

$$\mathbb{H}_h = \{\mathbf{u}_h\in\mathbb{U}_h \mid (\mathbf{u}_h, \mathrm{D_h}\phi_h) = 0, \forall\,\phi_h\in\mathbb{V}_h \text{ and } (\mathbf{u}_h, \nabla^\perp\gamma_h) = 0, \forall\,\gamma_h\in\mathbb{W}_h\}. \tag{44}$$

Now we characterize the discrete harmonic space, which has zero discrete divergence and zero weak curl, with the help of the discrete Helmholtz decomposition for BDM_k . First, from the theory of finite element exterior calculus [32], we know that there is a discrete Helmholtz decomposition for BDM_k, for any $\mathbf{u}_h \in \mathrm{BDM}_k$, there exist $\Phi_h \in \mathrm{DG}_{k-1}$, $\Psi_h \in \mathrm{P}_{k+1}$ and $\Theta_h \in \mathbb{H}_h^0$, such that

$$\mathbf{u}_h = \nabla\Phi_h + \nabla \times \Psi_h + \Theta_h, \tag{45}$$

where $\mathbb{H}_h^0$ is the space of two constant vectors in the periodic plane case. On the other hand, we notice that the kernel space of discrete divergence operator on $\mathbb{U}_h$ is exactly the kernel space of divergence on BDM_k [24, Lemma 3.3], hence from the definition of the Helmholtz decomposition for SDG_k and BDM_k, it shows the spaces $\nabla^\perp\mathbb{W}_h \oplus \mathbb{H}_h$ and $\nabla^\perp\mathrm{P}_{k+1} \oplus \mathbb{H}_h^0$ are the same. Note that $\mathbb{W}_h = \mathrm{P}_{k+1}$, it then follows the harmonic vector space $\mathbb{H}_h = \mathbb{H}_h^0$, which consists two constant vectors in the periodic plane case. The discrete Helmholtz decomposition (43) is a counterpart of the continuous one.

Remark 3.4. The compatible finite element method [33] with mixed finite elements ensures that the divergence of the velocity space is exactly the height space. This property guarantees that the divergence operator in the distributed sense aligns with the pointwise divergence operator. Such compatibility is crucial for deriving important properties, such as the absence of spurious inertial oscillations and the conservation of potential enstrophy. In contrast, although the SDG discretization has a discrete counterpart to the continuous Helmholtz decomposition, it does not fit into discrete exact sequences. As a result, our proposed method may still have spurious inertial oscillations and does not conserve the potential enstrophy exactly. Nevertheless, the primary focus of the proposed scheme is to enhance the efficiency of time marching.

3.5. Steady geostrophic mode

Large-scale flows in atmosphere and ocean models are energetically dominated by dynamics that nearly achieve hydrostatic and geostrophic balance, on the f-plane the geostrophic mode becomes steady and divergence free. It is crucial for numerical discretization methods to accurately represent these steady-state geostrophic modes. Failure to preserve this steady state, particularly when nonlinear dynamics and spherical geometries are factored into the model, can cause the balanced states to generate disruptive fast inertia-gravity waves, deteriorating the quality of numerical solutions.

Given any stream function $\Psi \in H_1(\Omega)$, let us set

$$\mathbf{u} = \nabla^\perp\Psi := \mathbf{k} \times (\nabla\Psi), \qquad h = (f/g)\,\Psi. \tag{46}$$

It is easy to check that

$$\nabla \cdot \mathbf{u} = 0, \qquad f\mathbf{u}^\perp + g\nabla h = 0,$$

and the above $(\mathbf{u}, h)$ defined by (46) results in a steady geostrophic balance system for the linearized shallow water equations (38)–(39). As shown below, our SDG discretization (40)–(41) can retain a discrete steady geostrophic mode corresponding to their continuous counterpart, achieved through proper projection of the continuous functions into the discrete finite element spaces. We have the following theorem.

Theorem 3.5. *There exists a steady-state solution to SDG discretization (40)–(41) that approximates the geostrophic mode (46) with optimal order.*

Proof. First we obtain a discrete stream function by the projection of Ψ onto space $\mathbb{W}_h$ as

$$\Psi_h = \Pi_{\mathbb{W}_h}\Psi.$$

Then we set the discrete velocity as

$$\mathbf{u}_h = \nabla^\perp\Psi_h, \tag{47}$$

(clearly $\mathbf{u}_h \in \mathbb{U}_h$ is normally continuous across all edges by Lemma 2.1) and set the discrete depth as

$$h_h = (f/g)\,\mathcal{I}_h\Psi_h, \tag{48}$$

where the interpolation operator $\mathcal{I}_h$ from $H^1(\Omega)$ to $\mathbb{V}_h$ is defined by: for each edge $e \in \mathcal{E}_{\mathrm{H1}}$

$$\int_e (\mathcal{I}_h h - h)\, p_k \,\mathrm{d}\sigma = 0, \quad \forall\, p_k \in P_k(e),$$

and for each element $\tau \in \mathcal{T}$

$$\int_\tau (\mathcal{I}_h h - h)\, p_{k-1} \,\mathrm{dS} = 0, \quad \forall\, p_{k-1} \in P_{k-1}(\tau).$$

The interpolator $\mathcal{I}_h$ is compatible in the sense that for any $\Psi \in H_1(\Omega)$,

$$\left(\mathcal{I}_h\Psi, \nabla \cdot \mathbf{v}_h\right) - \langle \mathcal{I}_h\Psi, [\![\mathbf{v}_h \cdot \mathbf{n}]\!]\rangle_{\mathcal{E}_{\mathrm{H1}}} = \left(\Psi, \nabla \cdot \mathbf{v}_h\right) - \langle \Psi, [\![\mathbf{v}_h \cdot \mathbf{n}]\!]\rangle_{\mathcal{E}_{\mathrm{H1}}}, \quad \forall\, \mathbf{v}_h \in \mathbb{U}_h, \tag{49}$$

and has optimal approximation properties [24]. Thus both $\mathbf{u}_h$ and h_h will converge to u and h with an optimal convergence rate related to the order of FEM spaces as the mesh refines.

Using the discrete divergence theorem (10), and note that geostatic $\mathbf{u}_h$ defined in (47) is continuous along normal direction across all triangle edges, and $\nabla \cdot \mathbf{u}_h = 0$, thus we have

$$\begin{aligned}\left(\frac{\partial h_h}{\partial t}, \phi_h\right) &= (H\mathbf{u}_h, \nabla\phi_h) - \langle H\mathbf{u}_h \cdot \mathbf{n}, [\![\phi_h]\!]\rangle_{\mathcal{E}_{\mathrm{Hdiv}}} \\ &= -(H\nabla\cdot\mathbf{u}_h, \phi_h) + \langle H[\![\mathbf{u}_h\cdot\mathbf{n}]\!], \phi_h\rangle_{\mathcal{E}_{\mathrm{H1}}} \\ &= 0, \qquad \forall\, \phi_h \in \mathbb{V}_h.\end{aligned} \tag{50}$$

Combining (41), (47), (49) and noticing that Ψ_h is continuous across all triangle faces, it holds

$$\begin{aligned}\left(\frac{\partial \mathbf{u}_h}{\partial t}, \mathbf{v}_h\right) &= -\left(f\mathbf{u}_h^{\perp}, \mathbf{v}_h\right) + \left(gh_h, \nabla\cdot\mathbf{v}_h\right) - \langle gh_h, [\![\mathbf{v}_h\cdot\mathbf{n}]\!]\rangle_{\mathcal{E}_{\mathrm{H1}}} \\ &= \left(f\nabla\Psi_h, \mathbf{v}_h\right) - \langle f[\![\Psi_h]\!], \mathbf{v}_h\rangle_{\mathcal{E}_{\mathrm{H1}}} + \left(gh_h, \nabla\cdot\mathbf{v}_h\right) - \langle gh_h, [\![\mathbf{v}_h\cdot\mathbf{n}]\!]\rangle_{\mathcal{E}_{\mathrm{H1}}} \\ &= \left(gh_h - f\Psi_h, \nabla\cdot\mathbf{v}_h\right) - \langle gh_h - f\Psi_h, [\![\mathbf{v}_h\cdot\mathbf{n}]\!]\rangle_{\mathcal{E}_{\mathrm{H1}}} \\ &= f\left(\mathcal{I}_h\Psi_h - \Psi_h, \nabla\cdot\mathbf{v}_h\right) - f\langle \mathcal{I}_h\Psi_h - \Psi_h, [\![\mathbf{v}_h\cdot\mathbf{n}]\!]\rangle_{\mathcal{E}_{\mathrm{H1}}} \\ &= 0, \qquad \forall\, \mathbf{v}_h \in \mathbb{U}_h.\end{aligned} \tag{51}$$

Therefore a discrete steady geostrophic mode $(\mathbf{u}_h, h_h)$ that approximates the continuous one (46) is acquired by (47)–(48).

4. Numerical dispersion analysis

The numerical dispersion analysis examines the modes in the discrete solution, that are expected to closely approximate the continuous modes. Failure to achieve this may result in spurious modes that are nonphysical.

Let us define the following two projections $P_{\mathbb{V}_h} : \mathbb{W}_h \to \mathbb{V}_h$ and $P_{\mathbb{W}_h} : \mathbb{V}_h \to \mathbb{W}_h$. For any $h_h \in \mathbb{V}_h$, define $P_{\mathbb{W}_h} h_h$ such that

$$(\nabla P_{\mathbb{W}_h} h_h, \nabla\gamma_h) = (\mathrm{D_h} h_h, \nabla\gamma_h), \qquad \forall\, \gamma_h \in \mathbb{W}_h, \tag{52}$$

which is uniquely defined by solving the Poisson equation on the periodic plane or sphere surface. For any $\zeta_h \in \mathbb{W}_h$, define $P_{\mathbb{V}_h}\zeta_h$ such that

$$(\mathrm{D_h} P_{\mathbb{V}_h}\zeta_h, \mathrm{D_h}\phi_h) = (\nabla\zeta_h, \mathrm{D_h}\phi_h), \qquad \forall\, \phi_h \in \mathbb{V}_h, \tag{53}$$

which corresponds to solving a Poisson equation using SDG method, i.e., find $(\hat{\mathbf{u}}_h, P_{\mathbb{V}_h}\zeta_h) \in \mathbb{U}_h \times \mathbb{V}_h$ such that

$$\begin{aligned}(\hat{\mathbf{u}}_h, \mathbf{v}_h) - (P_{\mathbb{V}_h}\zeta_h, \nabla\cdot\mathbf{v}_h) + \langle P_{\mathbb{V}_h}\zeta_h, [\![\mathbf{n}\cdot\mathbf{v}_h]\!]\rangle_{\mathcal{E}_{\mathrm{H1}}} &= 0, && \forall\, \mathbf{v}_h \in \mathbb{U}_h, \\ (\nabla\cdot\hat{\mathbf{u}}_h, \phi_h) - \langle [\![\mathbf{n}\cdot\hat{\mathbf{u}}_h]\!], \phi_h\rangle_{\mathcal{E}_{\mathrm{H1}}} &= (\nabla\zeta_h, \mathrm{D_h}\phi_h), && \forall\, \phi_h \in \mathbb{V}_h.\end{aligned}$$

It is easy to check the first equation indicates $\hat{\mathbf{u}}_h = -\mathrm{D_h} P_{\mathbb{V}_h}\zeta_h$, the second equation shows $(-\hat{\mathbf{u}}_h, \mathrm{D_h}\phi_h) = (\nabla\zeta_h, \mathrm{D_h}\phi_h)$ for any $\phi_h \in \mathbb{V}_h$, together they form (53).

We remark that the gradient space and the curl space of $\mathbb{W}_h$ are orthogonal, in the sense that for any $\gamma_h, \psi_h \in \mathbb{W}_h$,

$$(\nabla\gamma_h, \nabla^{\perp}\psi_h) = 0. \tag{54}$$

However, a similar orthogonal property does not hold for the discrete gradient operator $\mathrm{D_h}$ of the space $\mathbb{V}_h$, nevertheless, for any $\phi_h, \alpha_h \in \mathbb{V}_h$ and $\gamma_h \in \mathbb{W}_h$, using (42), we have that

$$(\mathrm{D_h}\phi_h, \mathrm{D_h}^{\perp}\alpha_h) = (\mathrm{D_h}\phi_h - \nabla\gamma_h, \mathrm{D_h}^{\perp}\alpha_h) \tag{55}$$

with $\mathrm{D_h}^{\perp}\alpha_h := \mathbf{k}\times(\mathrm{D_h}\alpha_h)$, thus the discrete gradient space and curl space of $\mathbb{V}_h$ are approximately orthogonal.

4.1. Spurious mode on f-plane

Now we analyze the spurious mode on the f-plane, where the Coriolis parameter f is constant, i.e., $f \equiv f_0$. We consider the time harmonic solution $he^{-i\omega t}$ and $\mathbf{u}e^{-i\omega t}$ for the linearized shallow water equations (38)–(39), and have the following theorem on numerical mode.

Theorem 3.4. *Consider the time harmonic solution of the SDG discretization (40)–(41), using the Helmholtz decomposition (43) for $\mathbf{u}_h$, it holds that*

$$\begin{aligned}&\left((\omega^2 - f_0^2)h_h, \phi_h\right) - (gH\mathrm{D_h}h_h, \mathrm{D_h}\phi_h) - f_0 H\left((\mathrm{D_h}\Phi_h)^{\perp}, \mathrm{D_h}\phi_h\right) \\ &\qquad - \frac{if_0^2 H}{\omega}\left(\mathrm{D_h}\left(P_{\mathbb{V}_h}P_{\mathbb{W}_h}\Phi_h - \Phi_h\right), \mathrm{D_h}\phi_h\right) = 0, \qquad \forall\, \phi_h \in \mathbb{V}_h.\end{aligned} \tag{56}$$

Proof. Using the Helmholtz decomposition (43) for the trial function $\mathbf{u}_h$, taking the test function $\nabla^\perp\gamma_h$ with $\gamma_h \in \mathbb{W}_h$ and $\mathrm{D_h}\phi_h$ with $\phi_h \in \mathbb{V}_h$ in (41), and the test function $\phi_h \in \mathbb{V}_h$ in (40), and using the orthogonal relation (42), (44) and (54), we obtain

$$-i\omega(\nabla\Psi_h, \nabla\gamma_h) + \left(f_0 \mathrm{D_h}\Phi_h, \nabla\gamma_h\right) = 0, \qquad \forall\, \gamma_h \in \mathbb{W}_h, \tag{57}$$

$$\begin{aligned} -i\omega(\mathrm{D_h}\Phi_h, \mathrm{D_h}\phi_h) + \left(f_0(\mathrm{D_h}\Phi_h)^\perp - f_0\nabla\Psi_h, \mathrm{D_h}\phi_h\right) \\ -\left(gh_h, \nabla\cdot\mathrm{D_h}\phi_h\right) + \langle gh_h, [\![\mathbf{n}\cdot\mathrm{D_h}\phi_h]\!]\rangle_{\mathcal{E}_{\mathrm{H1}}} = 0, \qquad \forall\, \phi_h \in \mathbb{V}_h, \end{aligned} \tag{58}$$

$$-i\omega(h_h, \phi_h) + (H\nabla\cdot\mathrm{D_h}\Phi_h, \phi_h) - \langle H[\![\mathbf{n}\cdot\mathrm{D_h}\Phi_h]\!], \phi_h\rangle_{\mathcal{E}_{\mathrm{H1}}} = 0, \qquad \forall\, \phi_h \in \mathbb{V}_h. \tag{59}$$

Based on (57) and the projection $P_{\mathbb{W}_h}$ in (52), we have

$$-i\omega\Psi_h + f_0 P_{\mathbb{W}_h}\Phi_h = 0 \tag{60}$$

From (59), it holds

$$-i\omega(h_h, \phi_h) - (\mathrm{D_h}\Phi_h, H\mathrm{D_h}\phi_h) = 0, \qquad \forall\, \phi_h \in \mathbb{V}_h. \tag{61}$$

Then (56) is obtained with (58), (60) and (61).

In the f-plane numerical mode (56), the first two terms are the discrete inertia-gravity mode corresponding to the continuous case. The third term of (56) is the discretization error produced from the orthogonality of discrete gradient and discrete curl, which affects the accuracy of the discrete solution but does not compromise the convergence rate. The last term of (56) relates the discretization error originated from the space projection $P_{\mathbb{V}_h}P_{\mathbb{W}_h}$. In the case $\dim(\mathbb{W}_h) < \dim(\mathbb{V}_h)$, such projection will have nonzero kernel, which may lead to spurious modes. For example, the $\mathrm{P_1}$-$\mathrm{RT_0}$-$\mathrm{DG_0}$ scheme has $\dim(\mathbb{W}_h) : \dim(\mathbb{V}_h) \approx 1 : 2$ and possesses severe spurious mode problem [9]. As for the proposed SDG scheme, it holds $\dim(\mathbb{U}_h) = 2\dim(\mathbb{V}_h)$, and since the null space of curl operator on $\mathbb{W}_h$ contains only constant functions, and the null space of discrete gradient operator on $\mathbb{V}_h$ is the same, by the Helmholtz decomposition (43) we deduce

$$\dim(\mathbb{U}_h) = (\dim(\mathbb{W}_h) - 1) + (\dim(\mathbb{V}_h) - 1)$$

on the sphere surface, and consequently

$$\dim(\mathbb{W}_h) = \dim(\mathbb{V}_h) + 2. \tag{62}$$

Therefore, from a dimensional perspective, the projection $P_{\mathbb{V}_h}P_{\mathbb{W}_h}$ is not expected to introduce significant error. Nevertheless, additional analysis is still desired to thoroughly evaluate its impact.

4.2. Spurious mode on β-plane

Next we analyze the dispersion relation on the β-plane, where the Coriolis parameter $f = f_0 + \beta y$. On a midlatitude β-plane, the planetary wave motion is governed by the linearized shallow Eqs. (38)–(39). Asymptotic expansion based on the Rossby number provides a systematic framework for analyzing fluid motion in a rotating system [30], and the β-plane analysis has been carried out for various types of FEMs in [11] and [6]. The Rossby number is defined as $Ro = U/fL$, where U is the characteristic velocity scale of the flow and L is the characteristic length scale. When the Rossby number is very small ($Ro \ll 1$), the Coriolis force dominates over inertial forces, and the system is approximately in geostrophic balance. The dominant terms in (38)–(39), $H\nabla\cdot\mathbf{u}$ and $f_0\mathbf{u}^\perp + g\nabla h$, correspond to such steady geostrophic mode, while the smaller terms involving time derivatives $\frac{\partial h}{\partial t}$, $\frac{\partial \mathbf{u}}{\partial t}$ and β terms $\beta\mathbf{u}^\perp$ dictate the mode's evolution. The continuous level analysis shows the β-plane mode satisfies

$$i\omega\left(h - \lambda^2\Delta h\right) + \beta\lambda^2 h_x = 0, \tag{63}$$

where $\lambda = \frac{\sqrt{gH}}{f_0}$ is the deformation radius. We have similar results on numerical mode on β-plane.

Theorem 4.2. *Suppose the terms $\frac{\partial h}{\partial t}$, $\frac{\partial \mathbf{u}}{\partial t}$ and $\beta\mathbf{u}^\perp$ are of smaller order in (40)–(41), and $\mathbf{u}_g = \mathrm{D_h}\Phi_g + \nabla^\perp\Psi_g$ is the leading order expansions of $\mathbf{u}_h = \mathrm{D_h}\Phi_h + \nabla^\perp\Psi_h$, then with the first order approximation of (40)–(41), we have numerical mode satisfies*

$$i\omega\left((\nabla\Psi_g, \nabla\gamma_h) + \left(\frac{f_0^2}{gH}P_{\mathbb{V}_h}\Psi_g, P_{\mathbb{V}_h}\gamma_h\right)\right) + \left(\beta\frac{\partial\Psi_g}{\partial x}, \gamma_h\right) = 0, \qquad \forall\, \gamma_h \in \mathbb{W}_h. \tag{64}$$

Proof. Again we apply the Helmholtz decomposition (43) for the trial function $\mathbf{u}_h$ in the SDG discretization (40)–(41) (with $\Theta_h = 0$ since none such mode exists on the sphere), and use the test function $\nabla^\perp\gamma_h$ with $\gamma_h \in \mathbb{W}_h$ in (41) and the test function $\phi_h \in \mathbb{V}_h$ in (40) to obtain

$$-i\omega(\nabla\Psi_h, \nabla\gamma_h) + \left((f_0 + \beta y)\mathrm{D_h}\Phi_h + (f_0 + \beta y)\nabla^\perp\Psi_h, \nabla\gamma_h\right) = 0, \qquad \forall\, \gamma_h \in \mathbb{W}_h, \tag{65}$$

$$-i\omega(h_h, \phi_h) + (H\nabla\cdot\mathrm{D_h}\Phi_h, \phi_h) - \langle H[\![\mathbf{n}\cdot\mathrm{D_h}\Phi_h]\!], \phi_h\rangle_{\mathcal{E}_{\mathrm{H1}}} = 0, \qquad \forall\, \phi_h \in \mathbb{V}_h. \tag{66}$$

We first determine the zeroth order expansions (65) and (66), representing the quasi-geostrophic limit, which satisfy $\nabla\cdot\mathbf{u} = 0$ and $f_0\mathbf{u}^\perp + g\nabla h = 0$. In our SDG discretization, it corresponds to

$$\Phi_g = 0, \qquad \left(f_0\nabla\Psi_g, \mathrm{D_h}\phi_h\right) - \left(g\mathrm{D_h}h_g, \mathrm{D_h}\phi_h\right) = 0, \qquad \forall\, \phi_h \in \mathbb{V}_h, \tag{67}$$

where $\mathbf{u}_g = \mathrm{D_h}\Phi_g + \nabla^\perp \Psi_g$ and h_g is the leading order terms in the low Rossby number expansions of $\mathbf{u}_h = \mathrm{D_h}\Phi_h + \nabla^\perp \Psi_h$ and h_h respectively. The equations (67) and (53) imply

$$h_g = f_0/g P_{\mathbb{V}_h} \Psi_g. \tag{68}$$

Then by applying the first order expansions (65) and (66), and using the definition of discrete gradient (34), we obtain

$$-i\omega(\nabla\Psi_g, \nabla\gamma_h) + \left(f_0 \mathrm{D_h}\Phi_h, \nabla\gamma_h\right) + \left(\beta y \nabla^\perp \Psi_g, \nabla\gamma_h\right) = 0, \qquad \forall\, \gamma_h \in \mathbb{W}_h, \tag{69}$$

$$-i\omega(h_g, \phi_h) - (H \mathrm{D_h}\Phi_h, \mathrm{D_h}\phi_h) = 0, \qquad \forall\, \phi \in \mathbb{V}_h. \tag{70}$$

Note that, unlike the f-plane analysis, only two equations are needed to solve the unknown Ψ_g. Using (70), (68) and (53), the second term of (69) becomes

$$\begin{aligned} \left(f_0\, \mathrm{D_h}\Phi_h, \nabla\gamma_h\right) &= \left(f_0\, \mathrm{D_h}\Phi_h, \mathrm{D_h} P_{\mathbb{V}_h}\gamma_h\right) \\ &= -i\omega\left(\frac{f_0}{H}\, h_g, P_{\mathbb{V}_h}\gamma_h\right) \\ &= -i\omega\left(\frac{f_0^2}{gH}\, P_{\mathbb{V}_h}\Psi_g, P_{\mathbb{V}_h}\gamma_h\right), \end{aligned} \tag{71}$$

and the third term of (69) becomes

$$\begin{aligned} \left(\beta y \nabla^\perp \Psi_g, \nabla\gamma_h\right) &= -\left(\nabla\cdot(\beta y \nabla^\perp \Psi_g), \gamma_h\right) + \langle [\![y\mathbf{n}\cdot\nabla^\perp\Psi_g]\!], \gamma_h\rangle_{\mathcal{E}_{\mathrm{H1}}} \\ &= -\left(\beta \frac{\partial \Psi_g}{\partial x}, \gamma_h\right) \end{aligned} \tag{72}$$

insert (71) and (72) into (69), and (64) is obtained.

The Eq. (64) corresponds to the continuous mode (63). However, the mass term

$$\left(\frac{f_0^2}{gH}\, P_{\mathbb{V}_h}\Psi_g, P_{\mathbb{V}_h}\gamma_h\right)$$

is only an approximation of $\left(\frac{f_0^2}{gH}\, \Psi_g, \gamma_h\right)$, which still could introduce spurious mode if the projection $P_{\mathbb{V}_h}$ has large kernel space. As discussed previously, the proposed SDG scheme possesses a favorable dimension configuration (62), mitigating this kernel problem to some extent.

5. Numerical experiments

Our SDG method will be validated against several standard benchmark tests [34,35] to assess its convergence and conservation properties. All simulations employ a third-order Strong Stability Preserving Runge-Kutta (SSP-RK3) scheme for time stepping [36], which is an explicit time stepping scheme and highly efficient when used with diagonal or block diagonal mass matrices. It is important to note that the discrete global energy conservation property established in (29) is for the continuous-time space-discrete setup. After the time discretization, since the time stepping SSP-RK3 is non-conservative, it will introduce numerical errors and slightly compromise the energy conservation. Nevertheless, the energy conservation can still be preserved up to the temporal order of the explicit integrator. On the other hand, implicit time schemes, such as time-centered implicit Poisson integrators based on the discrete gradient method [37,38], are capable of conserving energy both temporally and spatially. The time step size is determined according to a CFL type condition for explicit RKDG schemes,

$$\Delta t \le \frac{C_{\mathrm{CFL}}\, h_{\min}}{(2p+1)\, \lambda_{\max}}$$

where p is the polynomial order of the SDG function spaces and $\lambda_{\max} = |\mathbf{u}| + \sqrt{gh}$ is the maximum characteristic speed for the shallow water equation. Since the central fluxes are employed, the CFL number must be reduced compared to upwind schemes due to the lack of numerical dissipation, we use $C_{\mathrm{CFL}} = 0.25$ in our simulations. The polygonal meshes used by our SDG scheme will be generated using the Spherical Centroidal Voronoi Tessellation (SCVT) [39,40], a specialized form of Voronoi tessellation where the generating point of each Voronoi cell coincides with its centroid. The polygonal mesh is then divided into triangles to form primal and dual MEs for our SDG discretization. The normalized errors for the height and velocity fields are defined as in [34]:

$$e_{L_2}(u) = \frac{||u - u_{\mathrm{ref}}||_{L_2}}{||u_{\mathrm{ref}}||_{L_2}}, \qquad e_{L_\infty}(u) = \frac{||u - u_{\mathrm{ref}}||_{L_\infty}}{||u_{\mathrm{ref}}||_{L_\infty}}, \tag{73}$$

where u is the field in question and f_{ref} denotes the true solution.

5.1. Solid body rotation (the Williamson test case 2)

The test case 2 - solid body rotation by Williamson [34] describes a zonally symmetric, geostrophically balanced flow to the shallow-water equations. The stable equilibrium solution of this test is theoretically expected to be maintained exactly over time,

Table 1
The staggered grids and the discretization settings for the Williamson test case 2. The numbers of degrees of freedom (DOFs) for depth and velocity are provided along with the time step sizes for the SDG_1 and SDG_2 schemes.

Grid	Nodes	Triangles	Resolu.	SDG_1			SDG_2		
			(km)	DOFs(h)	DOFs(**u**)	Δt (s)	DOFs(h)	DOFs(**u**)	Δt (s)
R1	362	720	1380	1,440	2,880	960	3,240	6,480	480
R2	1,442	2,880	700	5,760	11,520	480	12,960	25,920	240
R3	5,762	11,520	350	23,040	46,080	240	51,840	103,680	120
R4	23,042	46,080	175	92,160	184,320	120	207,360	414,720	60

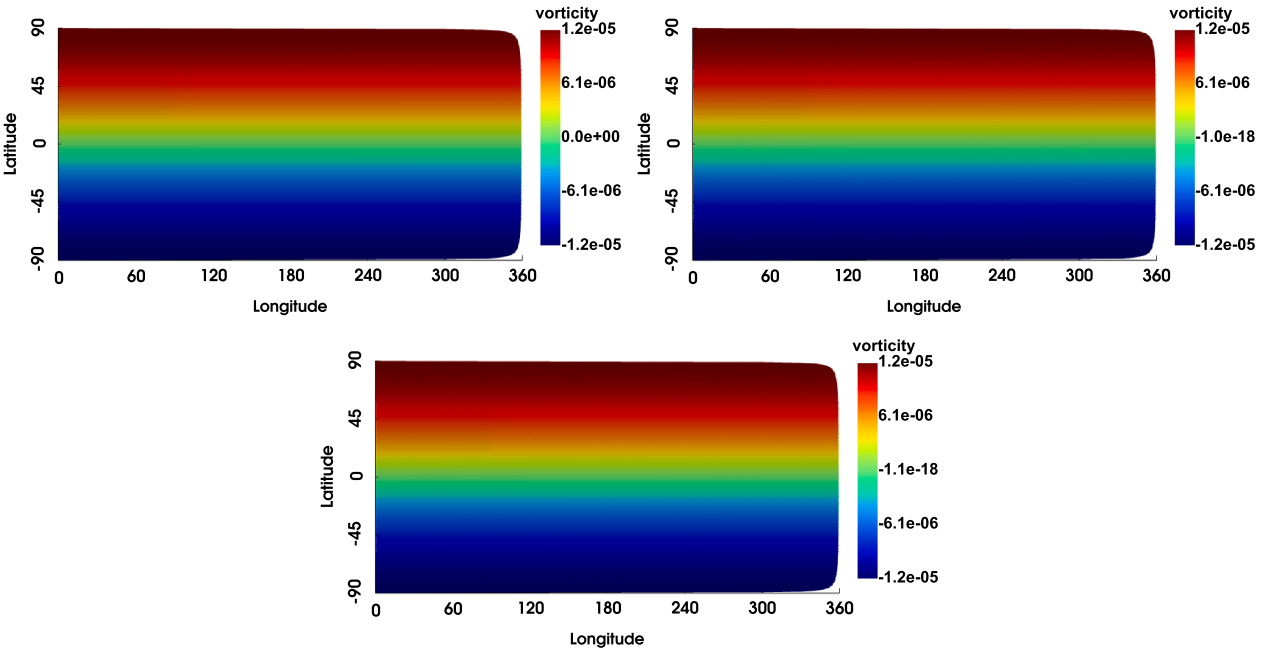


Fig. 4. The simulated vorticity fields by the SDG_2 scheme on the grid R_4 for the Williamson test case 2 at days 0 (top-left), 5 (top-right), and 15 (bottom).

which provides a complex test for conservative schemes capable of preserving geostrophic balance in the linearized shallow-water. Furthermore, the analytical solution is used to validate the convergence of the model. The initial height field of this test is set to be

$$h = h_0 - \left(a\omega u_0 + \frac{u_0^2}{2} \right) \frac{z^2}{ga^2}, \tag{74}$$

where $a = 6.37122 \times 10^6$ m is the radius of the Earth, $\omega = 7.29212 \times 10^{-5}$ s^{-1} is the angular speed of the Earth, $g = 9.80616$ m s^{-2} is the gravitational acceleration, and $h_0 = 2.94 \times 10^4$ m^{2}s$^{-2}/g$. The divergence free velocity field is given by

$$\mathbf{u} = \frac{u_0}{a}(-y, x, 0), \tag{75}$$

where $u_0 = 1\pi a/6$ (days)$^{-1}$. The SDG_1 and SDG_2 discretization are employed in this test case, the staggered grids and the discretization settings are listed in Table 1.

The equilibrium solution is expected to remain steady over time, therefore, we conducted numerical simulations for 15 days, and the numerical vorticity fields at day 0, 5 and 15 (produced by the SDG_2 scheme on the grid R_4) is shown in Fig. 4. The normalized L_2 errors for the four sets of staggered grids (R1, R2, R3 and R4) using the SDG_1 and SDG_2 schemes are presented in Fig. 5. The changes of numerical errors during the whole simulation period are negligible except the case of the coarsest grids in which there are some little oscillations in **u**. The convergence rate of L_2 and L_∞ error are presented in Fig. 6. Numerical results demonstrate that both L_2 and L_∞ normalized errors for height and velocity converge at a rate of $\mathcal{O}(\Delta x^2)$ for the SDG_1 scheme, and $\mathcal{O}(\Delta x^3)$ for the SDG_2 scheme, which are considered optimal. In scenarios with smooth solutions, as in this test case, the higher-order scheme shows greater efficiency. For instance, to achieve an L_2 error of height around 3×10^{-4}, the SDG_1 scheme requires the use of the grid R4 with 92,160 DOFs for the height variable and the time step size $\Delta t = 120$, whereas the SDG_2 scheme only needs the grid R2 with 12,960 DOFs for the height variable and the time step size $\Delta t = 240$. The conservation errors for mass, energy, and potential enstrophy are normalized

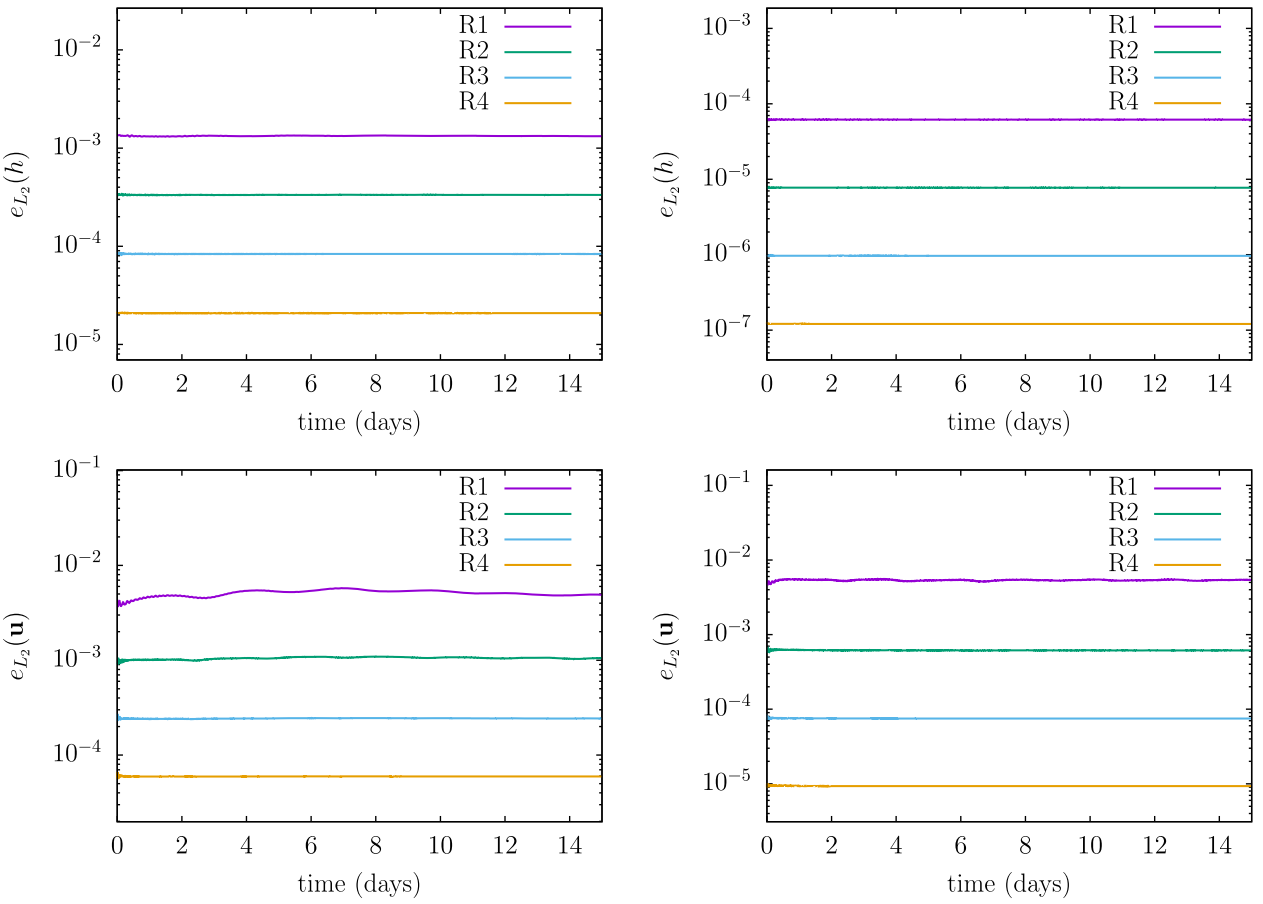


Fig. 5. The normalized L_2 errors in height and velocity for the Williamson test case 2. The left column shows results produced from the SDG_1 scheme and the right column shows results produced from the SDG_2 scheme. The top row shows the fluid height errors and the bottom row shows the velocity errors.

by their respective initial values, while the total vorticity is normalized as follows:

$$Q = \frac{\int_\Omega q_h h_h \, \mathrm{d}V}{\left\| q_h^0 \right\|_{L^2} \left\| h_h^0 \right\|_{L^2}}$$

where q_h^0 and h_h^0 are the initial values of potential vorticity and depth, respectively. Such a normalization approach follows that used in [41]. Given that the non-conservative SSP-RK3 time-stepping scheme is employed, the discrete energy is not exactly conserved, and the conservation error is related to the accuracy of the time-stepping scheme. In this test case, the grids and time step sizes listed in Table 1 are used to evaluate the normalized energy conservation errors. It is observed that as the grid and time step size are refined, the convergence rate is approximately 4, as shown in Fig. 7-(left). On the R4 grid, the normalized conservation errors of mass, energy, vorticity and potential enstrophy are shown in Fig. 7-(right). Both the mass conservation error and the vorticity conservation error remain very small, with the former exhibiting an accumulation of round-off errors. The solution reaches a nearly steady state after a short period, with the errors in energy conservation and potential enstrophy conservation remaining on the order of 10^{-11}.

To demonstrate the efficiency of our SDG schemes for time advancing, we compare the per RK substep time costs of the RT_k/DG_k and SDG_k schemes for first and second orders, and the result is reported in Table 2. For RT_1 and RT_2 FEM schemes, the mass matrix is solved using a Jacobi-preconditioned CG method, requiring 14 and 21 iterations to reach a relative tolerance of 10^{-12} respectively, whereas the SDG schemes use purely local mass inversion. The tests are performed on grid R4 with 4 processors for first order schemes and 16 processors for second order schemes. Updating $\mathbf{u}$ or $\mathbf{F}$ in SDG_k takes only 24 % (first order) and 16 % (second order) of the corresponding RT_k time, while the total substep times are reduced to 31 % and 21 %, respectively, confirming the high efficiency of our proposed SDG method.

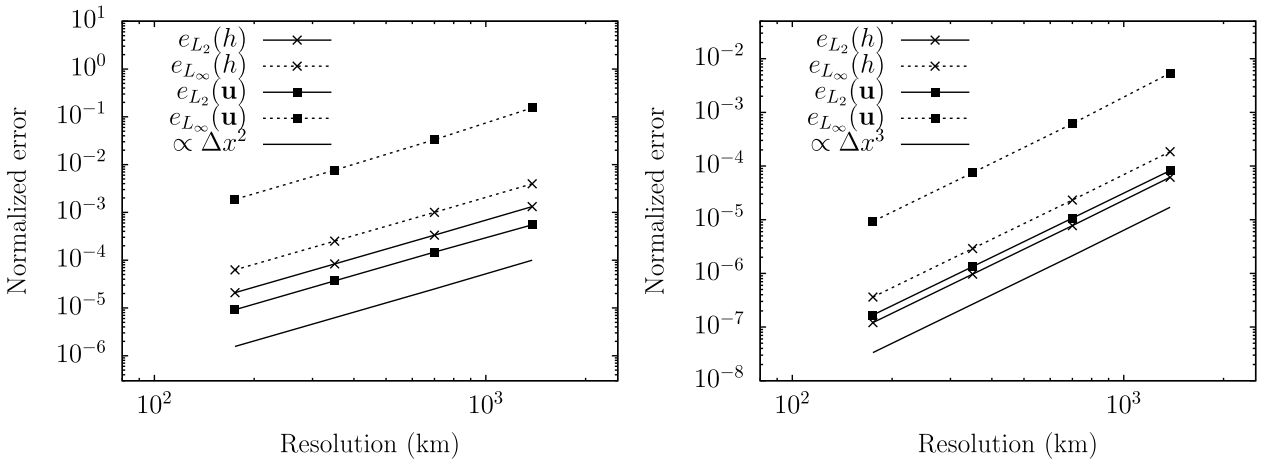


Fig. 6. The normalized L_2 and L_∞ errors of the SDG_1 (left) and SDG_2 (right) schemes for the Williamson test case 2 at day 15.

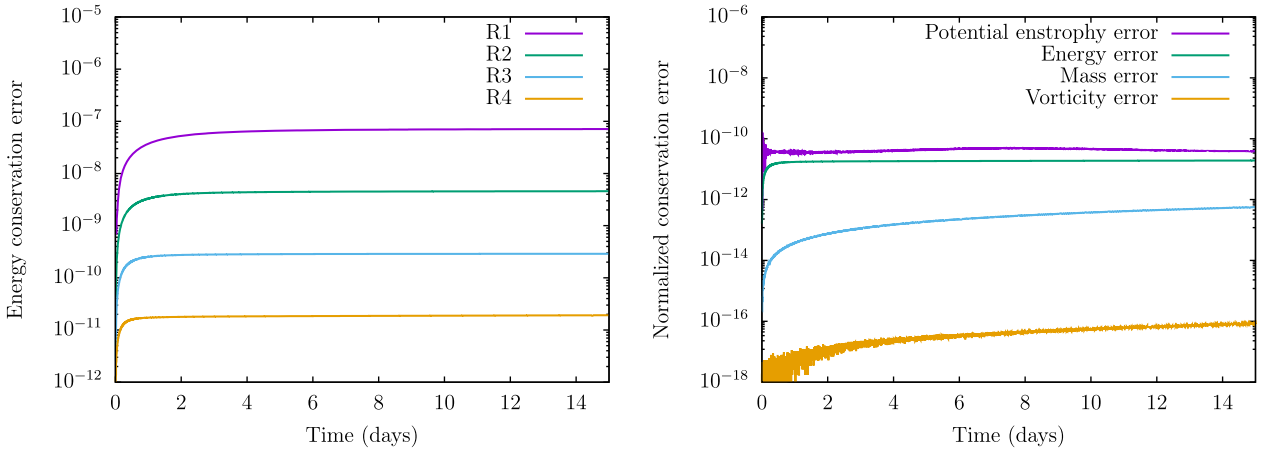


Fig. 7. The normalized conservation errors by the SDG_1 scheme for the Williamson test case 2. Left: The normalized conservation errors of energy with four grids and corresponding time step sizes. Right: The normalized conservation errors of mass, energy, vorticity and potential enstrophy on the grid R4.

Table 2
Breakdown and comparisons of the per RK substep time costs (ms) of the RT_k/DG_k and SDG_k schemes of first and second orders in the Williamson test case 2.

Variable updated	First order		Second order	
	RT_1/DG_1	SDG_1	RT_2/DG_2	SDG_2
F	29.40	7.02	29.69	4.79
u	30.52	7.16	28.04	4.75
U	1.20	1.22	0.50	0.51
h	1.40	1.31	0.82	0.91
ζ	1.85	2.03	0.53	0.74
q	1.01	1.22	0.28	0.53
Total	65.78	20.57	60.37	12.93

5.2. Flow over a mountain (the Williamson test case 5)

The Williamson test case 5 - flow over a mountain has a zonal flow with initial conditions similar to the test case 2, except $h_0 = 5960$ m and $u_0 = 20$ ms^{-1}. Additionally, it includes an isolated conical mountain given by

$$h_s = h_{s_0}\left(1 - r/R_0\right)$$

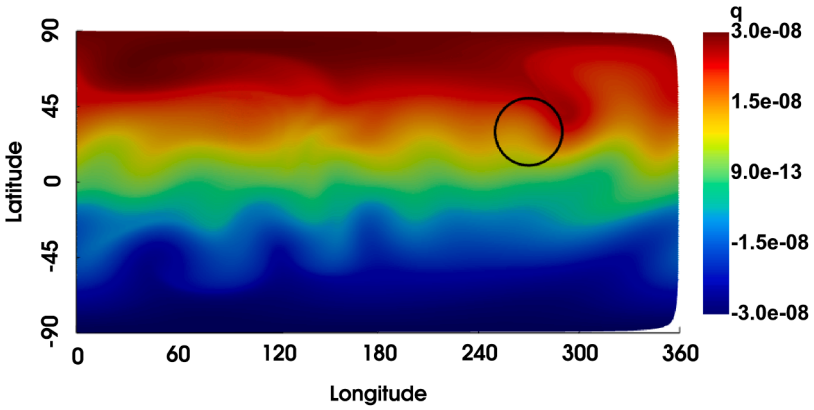


Fig. 8. The simulated potential vorticity for the Williamson test case 5 at day 15 by the SDG_1 scheme on the grid R4. The black circle marks the position of the mountain of the test.

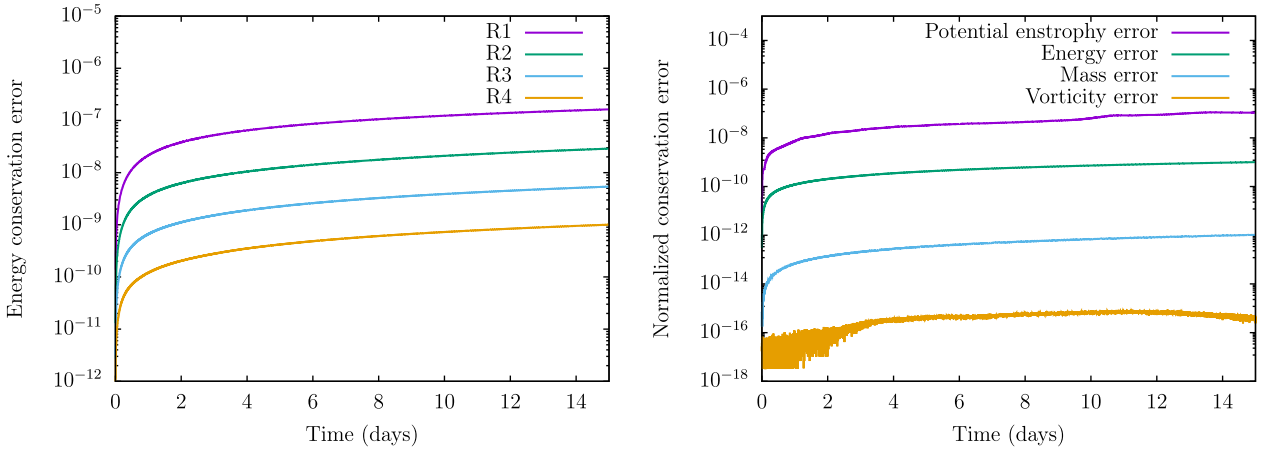


Fig. 9. The normalized conservation errors by the SDG_1 scheme for the Williamson test case 5. Left: The normalized conservation errors of energy with four grids and corresponding time step sizes. Right: The normalized conservation errors of mass, energy, vorticity and potential enstrophy on the grid R4.

where $h_{s_0} = 2000$ m, $R_0 = \pi/9$ and $r^2 = \min\left\{R_0^2,\ (\phi-\phi_c)^2 + (\lambda-\lambda_c)^2\right\}$ and the mountain is centered at $\phi_c = \pi/6$ and $\lambda_c = -\pi/2$. The fluid height is obtained by subtracting the topographic height h_{s_0} from the total depth h_0. As the geostrophically-balanced zonal flow interacts with the forcing mountain, it produces large amplitude Rossby and gravity waves that travel around the globe.

In this test case, the SDG_1 scheme and the grid R4 were used with a time step size $\Delta t = 60$ s. The simulation results after 15 days are presented in Fig. 8, which align well with the results reported in the literature [41]. The normalized conservation errors for this test are presented in Fig. 9. In this test case, we use the four set of grids R1 to R4 and corresponding time step sizes 480s, 240s, 120s and 60s to test the normalized energy conservation errors, which exhibit a convergence rate of 2.5 as the grid and time step size are refined. The normalized conservation errors for mass, energy, vorticity and potential enstrophy on the R4 grid are shown in Fig. 9. Again, the mass conservation error and the vorticity conservation error remain very small, the conservation error of potential enstrophy is two orders of magnitude larger than that of energy, since the proposed scheme does not preserve potential enstrophy.

5.3. *Barotropically unstable jet flow*

The test of barotropically unstable jet flow, as detailed by Galewsky in [35], focuses on the nonlinear barotropic instability of a mid-latitude jet. This test begins with a jet localized between 25°N and 65°N, traveling at approximately 80 m/s. The height field, geostrophically balanced with the jet, is modified by imposing a shallow Gaussian hill perturbation. This perturbation evolves by interacting with the mean flow, triggering a gravity wave that radiates outward and excites a shear instability, ultimately leading to turbulent decay. This test poses significant challenges for numerical methods due to potential instabilities from numerical imbalances or non-orthogonal grid alignments, making it a crucial benchmark for unstructured grids that may affect the instability's development and outcome.

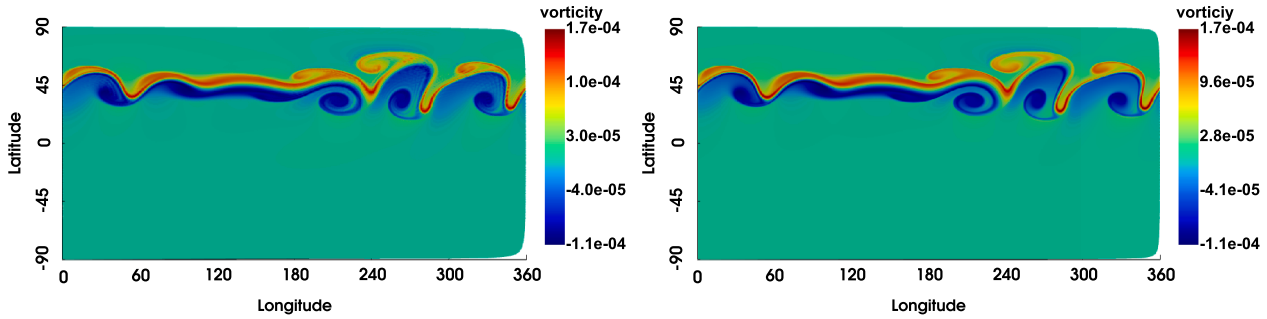


Fig. 10. The simulated normalized vorticity fields at day 6 for the barotropically unstable jet flow test by Galewsky using the SDG_1 scheme on the grid R4 (top) and the uniformly refined grid of R4 (bottom).

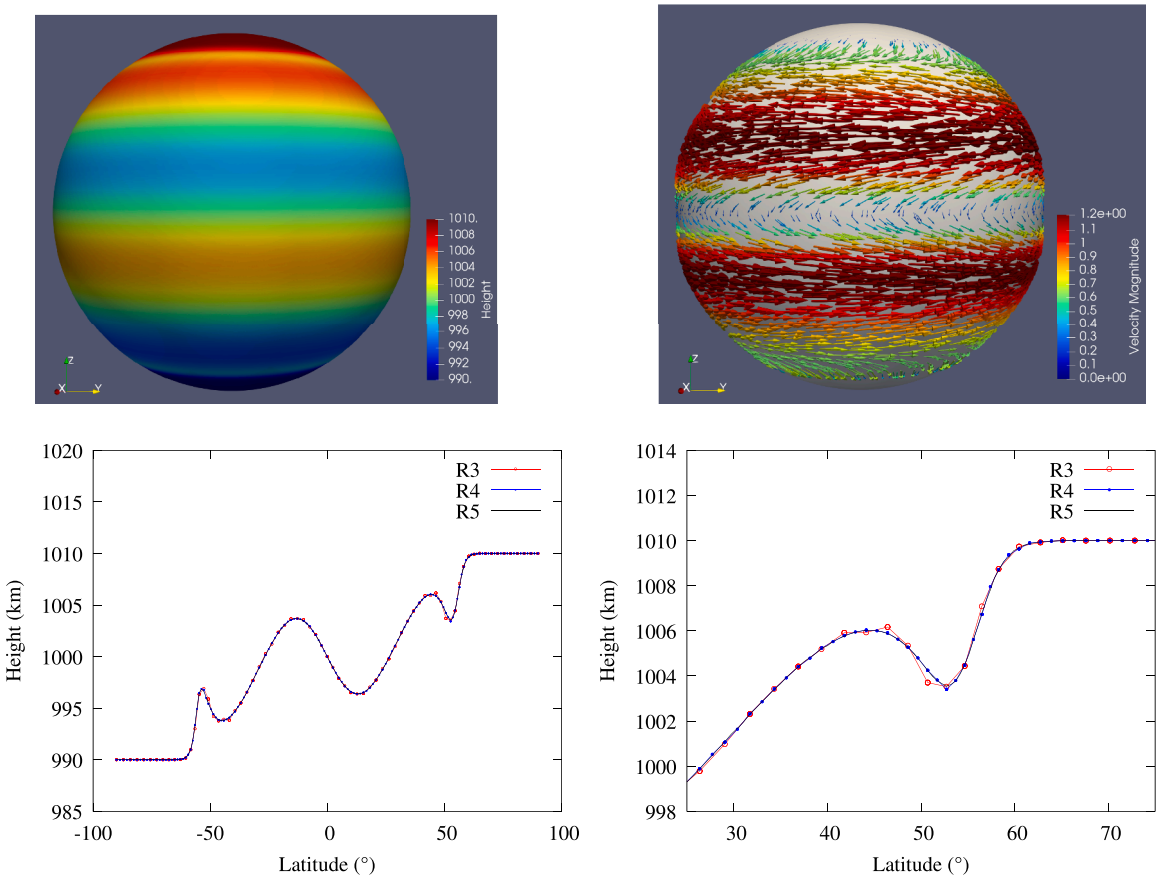


Fig. 11. Simulation results at $t = 62{,}400$ s (≈ 0.72 days) for the regularized dam break test case by the SDG_1 scheme. Top-left: the height on the grid R4; Top-right: the velocity on the grid R4; Bottom-left: the meridional section $h(\phi)$ at the longitude $\lambda = 0°$, Bottom-right: the same section, zoomed to $\phi \in [25°, 75°]$.

We employed the SDG_1 scheme on the grid R4 with the time step size $\Delta t = 60$ s and the uniformly refined grid of R4 with the time step size $\Delta t = 30$ s. We apply a biharmonic viscosity term, $\Delta^2\mathbf{u}$, with a coefficient of 8×10^{12} ms^{-1} in the velocity Eq. (2) to suppress grid-scale instabilities caused by nonlinear cascades, following the approach in [2,42]. This biharmonic viscosity operator induces energy dissipation, therefore the energy is not preserved in this case. The discretization of the biharmonic operator follows that of [12], and we omit the details here. The simulation results after 6 days are shown in Fig. 10, showing significant growth in the perturbation and a clear emergence of shear flow instability. Both numerical solutions produced by our model on the two different grids fall within the range of documented solutions in the literature [43,44], displaying the correct wavenumber and the expected quiescent section. The lower resolution simulation exhibits clear numerical dispersion, most notably in the vicinity of vortices, whereas

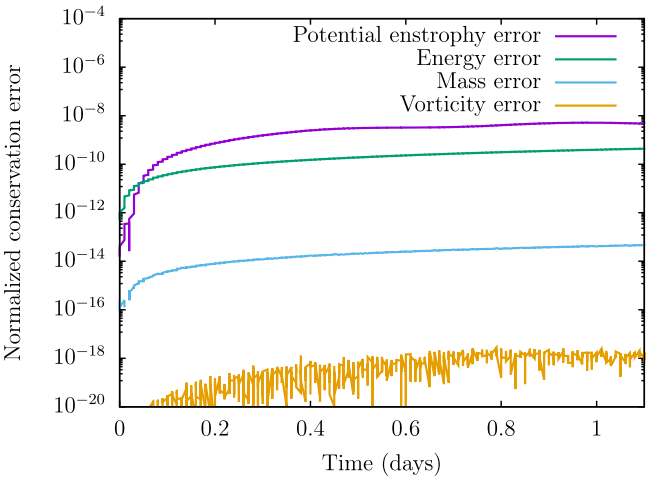


Fig. 12. The normalized conservation errors of mass, energy, vorticity and potential enstrophy on the grid R4 by the SDG_1 scheme for the regularized dam break test case.

this phenomenon is absent in the higher resolution simulation. This test case highlights the significance of higher resolution, allowing the model to accurately capture fine-scale features, minimize numerical artifacts, and improve the overall simulation accuracy.

5.4. Regularized dam break

We finally perform a regularized great circle dam break test, that imposes a thin tanh step across a great circle to generate bore-like yet continuous fronts on the sphere. This test targets the large scale geophysical regime, in which solutions are smooth and shocks are uncommon, and whether the numerical scheme can resolve steep gradients while preserving key invariants and early time front speeds. The initial velocity is zero and the initial thickness is defined as

$$h(\lambda, \phi, 0) = h_0 + \delta h \tanh\Big(\frac{a\phi}{d}\Big), \tag{76}$$

where ϕ denotes the latitude, $h_0 = 1000$ m, $\delta h = 10$ m (i.e., $\delta h/h_0 = 1\,\%$) and the transition length scale $d = 300$ km. The small amplitude of height perturbation keeps the flow in a smooth, large scale regime where the front remains continuous. We ran the test with the SDG_1 scheme on grids R3, R4, and the uniformly refined R4 so that the tanh transition is resolved by at least three grid cells on every level of grids. Fig. 11 shows some numerical simulation results. It is seen that at $t = 62{,}400$ s (≈ 0.72 days), two counter-propagating bore-like yet continuous fronts developed along the great circle, accompanied by a corresponding pair of Coriolis force induced zonal jets. Our scheme tracks the fronts cleanly, no spurious oscillations are observed even on the coarsest grid (R3), and the front positions follow the expected early time speed $c_0 = \sqrt{gh_0}$. Fig. 12 presents the normalized conservation errors of mass, energy, vorticity and potential enstrophy on the grid R4, demonstrating that the trends are consistent with the previous tests.

6. Conclusions

In conclusion, this paper presents a staggered discontinuous Galerkin method to address the shallow water equations on staggered triangular grids. The SDG discretization is carefully designed to preserve conservation properties, including local conservation of the mass and the global conservation of the energy and vorticity. Furthermore, by integrating SDG discretization with mass-lumping finite element methods for the vorticity equation, we achieve either block diagonal mass matrices for velocity and height equations and diagonal mass matrices for vorticity equation, which greatly improve the efficiency of explicit time-steppings. On the other hand, potential enstrophy is not theoretically conserved in our SDG schemes. In the cases of two dimensional turbulent fluids, potential enstrophy cascades to smaller scales and requires a dissipation mechanism, therefore, its conservation is less critical than that of energy. In this study we have utilized biharmonic dissipation to suppress the nonlinear cascades to grid scale, the other dissipation mechanisms such as the Anticipated Vorticity Method (APVM) and streamline-upwind Petrov-Galerkin (SUPG) are also worth further exploring. The global energy preservation in this study is achieved only in the continuous-time setup, and non-conservative time stepping schemes might compromise such conservation at the time-discrete level, developing fully discrete schemes with structure-preserving time matching method will be a focus in our future work. Moreover, while the present work focuses on smooth, large-scale geophysical flows, the inviscid shallow water equations can also develop discontinuities such as hydraulic jumps and bores. Extending the SDG framework with shock capturing mechanisms, such as subcell finite volume limiters, artificial viscosity, or WENO reconstructions, would broaden its applicability to extreme events including dam break problems and tsunamis. Incorporating such techniques, while retaining the conservation and efficiency properties of the SDG scheme, will be a natural and important direction of our future work.

CRediT authorship contribution statement

Wei Leng: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization; **Rihui Lan:** Writing – review & editing, Software, Methodology, Investigation, Funding acquisition, Formal analysis; **Lili Ju:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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